

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A150HAP10HP001Continuous Blowdown Drum - 7 Percent.pvdb

Date of Analysis : Sep 17, 2026 5:36pm

PV Elite 28 U1, March 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 550.000 mm. ID
 Vessel Design Length, Tangent to Tangent 2475.00 mm.
 Distance of Bottom Tangent above Grade 300.00 mm.
 Specified Datum Line Distance 50.00 mm.
 Internal Design Temperature 155 °C
 Internal Design Pressure 3.500 bars
 External Design Temperature 155 °C
 External Design Pressure 1.034 bars
 Maximum Allowable Working Pressure 4.12 bars
 External Max. Allowable Working Pressure 5.91 bars
 Shop Test Pressure 5.358 bars
 Field Test Pressure 5.358 bars
 Required Minimum Design Metal Temperature -10.0 °C
 Warmest Computed Minimum Design Metal Temperature -48.0 °C
 Wind Design Code ASCE-2010
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Nozzle	SA-105	...	...	K03504	
Nozzle	SA-106 B	...	...	K03006	
Nozzle Flg	SA-105	...	...	K03504	
Leg Baseplate		...	...		
Leg Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Bottom D end	3.623	1.03	29.800	3.0000	No	No
Shell	3.606	1.03	29.000	3.0000	N/A	No
Top D end	3.500	1.03	29.900	3.0000	No	No

Liquid Level: 1337.50 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 0.936

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Ellipse	0.0	50.0	12.0	10.0	4.5	4.5	0.85	1.00
Cylinder	2375.0	2375.0	10.0	10.0	4.5	6.4	0.85	1.00
Ellipse	2425.0	50.0	12.0	10.0	4.5	4.5	0.85	1.00

Local Stress Analysis Results:



Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
LEGS	WRC-107/537	0.173	n/a	Passed
LEGS	WRC-107/537	0.083	n/a	Passed
B2_2"	WRC-107/537	0.382	n/a	Passed
A1_4"	WRC-297	0.957	Shell	Passed
B1_3"	WRC-107/537	0.387	n/a	Passed
S1_8"	WRC-107/537	0.563	n/a	Passed

Loads for Foundation/Support Design:

Total Wind Shear on top of all Legs	259. Kg
Total Earthquake Shear on top of all Legs	167. Kg
Total Wind Moment at top of all Legs	269. Kg-m.
Total Earthquake Moment at top of all Legs	226. Kg-m.
Wind Load Case Test Moment on Support	89. Kg-m.
Test Case Wind Shear on Support	85. Kg
Max. Wind Shear on one Leg (top & bottom)	172. Kg
Max. Earthq. Shear on one Leg (top & bottom)	110. Kg
Max. Wind Moment at base of one Leg	246. Kg-m.
Max. Earthquake Moment at base of one Leg	158. Kg-m.
Max. Vertical Load (Wt. + Wind) on one Leg	1170. Kg
Max. Vertical Load (Wt. + Eq.) on one Leg	1039. Kg

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	765.3 kg.
Shop Test - Fabricated + water (full)	1396.6 kg.
Shipping - Fab. + rem. intls.+ shipping app.	799.4 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	799.4 kg.
Empty - Fab. + intls. + details + wghts.	799.4 kg.
Operating - Empty + operating liquid (no ca)	1086.5 kg.
Field Test - Empty weight + test medium (full)	1430.7 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
B2_2"	29.8	...	...	OK	4.50	OK	No Calc[*]
H1_6"	29	OK	...	...		OK	Passed
L2_2"	29	...	...	OK	4.50	OK	No Calc[*]
L4_2"	29	...	...	OK	4.50	OK	No Calc[*]
L1_2"	29	...	...	OK	4.50	OK	No Calc[*]
L3_2"	29	...	...	OK	4.50	OK	No Calc[*]
A1_4"	24.6	OK	...	OK	4.50	OK	Passed
A1_3"	29	...	...	OK	4.50	OK	No Calc[*]
B1_3"	15.1	OK	...	OK	4.50	OK	Passed
B1_3"	29.9	...	...	OK	4.50	OK	No Calc[*]
S1_8"	28.2	OK	...	OK	4.50	OK	Passed
S1_8"	29.2	OK	...	OK	4.50	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 15.1 Nozzle : B13"
 Minimum MAWP Shells/Flanges : 4.12 bars

Computed Vessel M.A.W.P. : 4.12 bars

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
10	B2_2"	0.000	0.000	113.600
20	H1_6"	400.000	0.000	316.800
20	L2_2"	650.000	0.000	113.600
20	L4_2"	650.000	90.000	113.600
20	L1_2"	1650.000	90.000	113.600
20	L3_2"	1650.000	0.000	113.600
20	A1_4"	1650.000	180.000	215.200
20	A1_3"	1650.000	180.000	150.862
30	B1_3"	0.000	0.000	164.400
30	S1_8"	0.000	180.000	418.400

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

Diameter Limit clash between : A13" and A14".
 The Centerline Dist. is 0.00 while the sum of the limit Radii is 183.03 mm..

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
B2_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	218.15	300
L2_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	222.31	150
L4_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	222.31	150
L1_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	222.31	150
L3_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	222.31	150
A1_3"	3.000 in	160	WNF	3.500	11.125	...	...	460.00	600
B1_3"	3.000 in	Actual	WNF	3.472	6.000	...	...	169.09	300
A1_4"	4.000 in	Actual	WNF	4.630	8.000	...	...	231.69	300
H1_6"	6.000 in	Actual	WNF	6.472	6.000	...	...	181.33	150
S1_8"	8.000 in	Actual	WNF	8.787	10.000	...	...	273.79	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
B2_2"	SA-105	10.000	10.000	...	...	...
L2_2"	SA-105	10.000	10.000	...	...	...
L4_2"	SA-105	10.000	10.000	...	...	...
L1_2"	SA-105	10.000	10.000	...	...	...
L3_2"	SA-105	10.000	10.000	...	...	...
A1_3"	SA-106 B	10.000	10.000	...	...	...
B1_3"	SA-105	10.000	10.000	...	...	...
A1_4"	SA-105	10.000	10.000	...	...	...
H1_6"	SA-105	10.000	10.000	...	...	...
S1_8"	SA-105	10.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
B2_2"	...	0.0	206.38	0.00	Bottom D end
L2_2"	600.000	0.0	209.55	0.00	Shell
L4_2"	600.000	90.0	209.55	0.00	Shell
L1_2"	1600.000	90.0	209.55	0.00	Shell
L3_2"	1600.000	0.0	209.55	0.00	Shell
A1_3"	1600.000	180.0	350.00	100.00	Shell
B1_3"	...	0.0	122.28	0.00	Top D end
A1_4"	1600.000	180.0	209.55	0.00	Shell
H1_6"	350.000	0.0	150.00	0.00	Shell
S1_8"	...	180.0	200.00	0.00	Top D end



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 12.0mm. THK X 550.0mm. ID X 50.0mm.	SA-516 70
2	INSULATION: 187mm X 30mm THK	HC
1	INSULATION: 2375mm X 30mm THK	HC
3	LEGS, SHAPE: L4X4X0.3125 1430.0mm	...
3	BASEPLATES: 154.0mm X 154.0mm X 16.0mm	SA-516 70
3	BASEPLATE BOLTS: 20.0mm	SA-36
3	LEG RE-PADS: 150.000mm X 150.000mm X 10.000mm	SA-516 70
1	CLASS 150 GR 1.1, 6.0" BLIND FLANGE(S)	SA-105
1	CLASS 150 GR 1.1, 2.0" BLIND FLANGE(S)	SA-105
1	CLASS 300 GR 1.1, 4.0" BLIND FLANGE(S)	SA-105
2	CYLINDRICAL SEGMENT 10.0mm. THK X 863.9mm. X 2375.0mm.	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Bottom D end	[10]	B	-29	-104	-29	0.182	10.000	0.85	No
Bottom D end	[7]	B	-23	-104	-29	0.095	12.000	0.85	No
Shell	[8]	B	-29	-104	-29	0.182	10.000	0.85	No
Top D end	[10]	B	-29	-104	-29	0.182	10.000	0.85	No
Top D end	[7]	B	-23	-104	-29	0.092	12.000	0.85	No
B22"	[1]	A	-8	-104	-29	0.093	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
H16"	[1]	A	-8	-104	-29	0.068	6.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
L22"	[1]	A	-8	-104	-29	0.102	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
L42"	[1]	A	-8	-104	-29	0.102	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
L12"	[1]	A	-8	-104	-29	0.101	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
L32"	[1]	A	-8	-104	-29	0.101	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
A14"	[1]	A	-8	-104	-29	0.027	8.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
A13"	[1]	B	-8	-104	-29	0.020	9.735	1.00	No
Nozzle Flg	[5]	A	-18	-104					
B13"	[1]	A	-8	-104	-29	0.035	6.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
S18"	[1]	A	-8	-104	-29	0.038	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Nozl. Bolting	[22]		-48			1.000	64.000	1.00	No

Warmest MDMT: -8 -104

Required Minimum Design Metal Temperature -10.0 °C
 Warmest Computed Minimum Design Metal Temperature -48.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and



UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note	10	20	The wind diameter multiplier is very big.
Note	20	30	The wind diameter multiplier is very big.
Note	30	40	The wind diameter multiplier is very big.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (barsUniCode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest) 3.5 bars
 Design Internal Temperature 155.0 °C
 Projection of Nozzle from Vessel Top 500 mm.
 Projection of Nozzle from Vessel Bottom 500 mm.
 Minimum Design Metal Temperature -10.0 °C
 Type of Construction Welded
 Special Service None
 Degree of Radiography RT-1
 Use Higher Longitudinal Stresses (Flag) Y
 Select t for Internal Pressure (Flag) N
 Select t for External Pressure (Flag) N
 Select t for Axial Stress (Flag) N
 Select Location for Stiff. Rings (Flag) N
 Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Field Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Vertical
 Test Performed in Corroded Condition Yes

Load Case 1 NP+EW+1.1WI+FW+BW
 Load Case 2 NP+EW+EE+FS+BS
 Load Case 3 NP+OW+1.1WI+FW+BW
 Load Case 4 NP+OW+EQ+FS+BS
 Load Case 5 NP+HW+HI
 Load Case 6 NP+HW+HE
 Load Case 7 IP+OW+1.1WI+FW+BW
 Load Case 8 IP+OW+EQ+FS+BS
 Load Case 9 EP+OW+1.1WI+FW+BW
 Load Case 10 EP+OW+EQ+FS+BS
 Load Case 11 HP+HW+HI
 Load Case 12 HP+HW+HE
 Load Case 13 IP+WE+EW
 Load Case 14 IP+WF+CW
 Load Case 15 IP+VO+OW



Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW
Wind Design Code	ASCE-7 2010
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Importance Factor	1.0
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 mm.
Distance Upwind of Crest	Lh 0 mm.
Distance from Crest to the Vessel	x 0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 0.000
Force Factor	0.000
Consider Vertical Acceleration	No
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR] 0.000
Component strength factor	[Rpo] 0.000
Overstrength factor	[Omega0] 0.000
Response Acceleration Factor	[SMS] 0.326
Response Acceleration Factor	[SM1] 0.139
Building Importance Factor	[Ie] 1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:



Element#	1/3	
Element From Node	10	
Element To Node	20	
Element Type	Elliptical	
Description	Bottom D end	
Distance "FROM" to "TO"	50	mm.
Inside Diameter	550	mm.
Element Thickness	10	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	12	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	3.5	bars
Design Temperature Internal Pressure	155	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	155	°C
Effective Diameter Multiplier	1.88	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm2
Allowable Stress, Operating	138	N./mm2
Allowable Stress, Hydrotest	234	N./mm2
Material Density	0.00775	kg./cm3
P Number Thickness	32.004	mm.
Yield Stress, Operating	231.2	N./mm2
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid 10	
Dist. from "FROM" Node / Offset dist	-137.5	mm.
Height/Length of Liquid	187.5	mm.
Liquid Density	0.0009347	kg./cm3
Element From Node	10	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	-137.5	mm.
Height/Length of Insulation	187.5	mm.
Thickness of Insulation	30	mm.
Density	0.0002	kg./cm3
Element From Node	10	
Detail Type	Nozzle	
Detail ID	B22"	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	10	
Detail Type	Weight	
Detail ID	Vortex breaker	
Dist. from "FROM" Node / Offset dist	-137	mm.
Miscellaneous Weight	2	Kgf
Offset from Element Centerline	0	mm.



Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	2375 mm.
Inside Diameter	550 mm.
Element Thickness	10 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	10 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	155 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	155 °C
Effective Diameter Multiplier	1.88
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1150 mm.
Liquid Density	0.0009347 kg./cm3

Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	2375 mm.
Thickness of Insulation	30 mm.
Density	0.0002 kg./cm3

Element From Node	20
Detail Type	Nozzle
Detail ID	H16"
Dist. from "FROM" Node / Offset dist	350 mm.
Nozzle Diameter	6 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	34.287 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	20
Detail Type	Nozzle
Detail ID	L22"
Dist. from "FROM" Node / Offset dist	600 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	20
Detail Type	Nozzle
Detail ID	L42"
Dist. from "FROM" Node / Offset dist	600 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	N



Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	L12"	
Dist. from "FROM" Node / Offset dist	1600	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	L32"	
Dist. from "FROM" Node / Offset dist	1600	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	A14"	
Dist. from "FROM" Node / Offset dist	1600	mm.
Nozzle Diameter	4	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	36.022	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	A13"	
Dist. from "FROM" Node / Offset dist	1600	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	20	
Detail Type	Leg	
Detail ID	LEGS	
Dist. from "FROM" Node / Offset dist	150	mm.
Diameter at Leg Centerline	648.82	mm.
Leg Orientation	3	
Number of Legs	3	
Section Identifier	L4X4X0.3125	
Length of Legs	1430	mm.
Element From Node	20	
Detail Type	Weight	
Detail ID	Lifting WT	
Dist. from "FROM" Node / Offset dist	1250	mm.



Miscellaneous Weight	25	Kgf
Offset from Element Centerline	0	mm.
Element From Node	20	
Detail Type	Weight	
Detail ID	insu. CLIP WT	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	31	Kgf
Offset from Element Centerline	0	mm.

Element#	3/3
Element From Node	30
Element To Node	40
Element Type	Elliptical
Description	Top D end
Distance "FROM" to "TO"	50 mm.
Inside Diameter	550 mm.
Element Thickness	10 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	155 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	155 °C
Effective Diameter Multiplier	1.88
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	187.5 mm.
Thickness of Insulation	30 mm.
Density	0.0002 kg./cm3

Element From Node	30
Detail Type	Nozzle
Detail ID	B13"
Dist. from "FROM" Node / Offset dist	200 mm.
Nozzle Diameter	3 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	14.025 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	30
Detail Type	Nozzle
Detail ID	S18"
Dist. from "FROM" Node / Offset dist	150 mm.
Nozzle Diameter	8 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	43.823 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Bottom D end		...	50	...	50
Shell		...	2425	...	2375
Top D end		...	2475	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Bottom D end		3.6227	12	3	550	117.3	1
Shell		3.6055	10	3	550	117.3	2
Top D end		3.5	12	3	550	117.3	3

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Bottom D end		3.5	29.8	42.5	10	4.5	1
Shell		3.5	29	41.7	10	4.5	2
Top D end		3.5	29.9	42.5	10	4.5	3

Minimum 4.12 19.6

Warning: A standard flange is governing the MAWP!

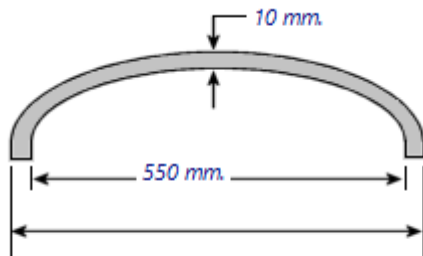
Warning: A standard flange is governing the MAP(NC)!

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Bottom D end

Elliptical Head Design Information:							
Design Pressure		P	3.623 bars	Int. Design Temperature		155 °C	
Material			SA-516 70	Ext. Design Temperature		155 °C	
Length			50.000 mm.	External Pressure Chart		CS-2	
				UNS Number		K02700	
Int. Corr. All.		c	3.0000 mm.	Allowable Stress (ope)		S	138.0 N./mm^2
Ext. Corr. All.		ce	0.0000 mm.	Allowable Stress (amb)		Sa	138.0 N./mm^2
Inside Diameter		Di	550.000 mm.	Actual Stress			16.7 N./mm^2
Spec. Min. Thk t _e , or sa			10.0000 mm.	Spec. Nominal Thk.			12.000 mm.
Surface Area			4473.1 cm^2	SG of Contents			0.936
Element Volume			33663.6 cm3	Weight of Contents			31.5 Kg
Empty Weight			43.4 Kg	Operating Weight			74.9 Kg
				Comp. All. Ope. Stress			133.0 N./mm^2
				Aspect Ratio			2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.875 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.438 mm.

Required Thickness due to Internal Pressure [t_r]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (3.62 \cdot 556 \cdot 0.99) / (2 \cdot 138 \cdot 0.85 - 0.2 \cdot 3.62)$$

$$= 0.8468 + 3.0000 = 3.8468 \text{ mm.}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm. per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.123 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 7) / (0.99 \cdot 556 + 0.2 \cdot 7)$$

$$= 29.9 - 0.12 = 29.8 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 10) / (1 \cdot 550 + 0.2 \cdot 10)$$

$$= 42.5 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [S_{act}]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (3.62 \cdot (0.99 \cdot 556 + 0.2 \cdot 7)) / (2 \cdot 0.85 \cdot 7)$$

$$= 16.730 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (3.62 \cdot 278) / (138 \cdot 0.85 - 0.6 \cdot 3.62) + 3$$

$$= 3.860 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.110 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 9) / (278 + 0.6 \cdot 9)$$

$$= 37.2 - 0.11 = 37.1 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (556 / (2 \cdot 140))^2) / 6$$

$$= 0.985841$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 9.045 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 10$, $t_r = 1.5$, $c = 3 \text{ mm.}$, $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.18$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 12$, $t_r = 1.01$, $c = 3 \text{ mm.}$, $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.095$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.244 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	10.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.182	Temperature Difference	75 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.875 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.438 mm.

EXTERNAL PRESSURE RESULTS for: Bottom D end

Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 155 °C

Elastic Modulus from Chart: CS-2 at 155 °C: 199186720 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
7.000	570.00	81.43	0.0017057	98.74



$$MAEP = B/(K0*Do/t) = 98.7/(0.9 *81.4) = 13.5 \text{ bars}$$

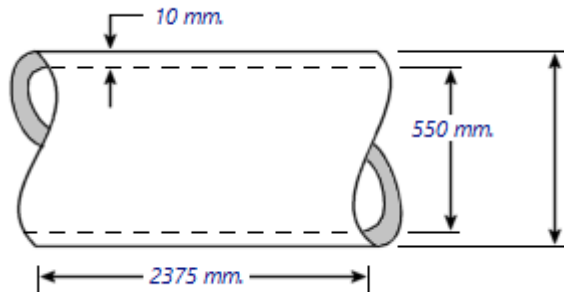
Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
1.479	570.00	385.49	0.0003603	35.88

$$MAEP = B/(K0*Do/t) = 35.9/(0.9 *385) = 1.03 \text{ bars}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell

Cylindrical Shell Design Information:					
Design Pressure	P	3.606 bars	Int. Design Temperature		155 °C
Material		SA-516 70	Ext. Design Temperature		155 °C
Length		2375.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	550.000 mm.	Actual Stress		17.1 N./mm^2
Spec. Min. Thk t _e , or sa		10.0000 mm.	Spec. Nominal Thk.		10.000 mm.
Surface Area		42529.3 cm^2	SG of Contents		0.936
Element Volume		564360.6 cm3	Weight of Contents		255.7 Kgf
Empty Weight		323.8 Kgf	Operating Weight		579.5 Kgf
			Comp. All. Ope. Stress		133.0 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3.61 \cdot 278) / (138 \cdot 0.85 - 0.6 \cdot 3.61)$$

$$= 0.8562 + 3.0000 = 3.8562 \text{ mm.}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm. per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.106 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 7) / (278 + 0.6 \cdot 7)$$

$$= 29.1 - 0.11 = 29 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 10) / (275 + 0.6 \cdot 10)$$

$$= 41.7 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (3.61 \cdot (278 + 0.6 \cdot 7)) / (0.85 \cdot 7)$$

$$= 17.102 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.786 %



Minimum Design Metal Temperature Results:

Govrn. thk, tg = 10, tr = 1.5, c = 3 mm., E* = 0.85
 Thickness Ratio = tr * E*/(tg - c) = 0.18, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.227 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	10.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.182	Temperature Difference	75 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: Shell

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 155 °C

Elastic Modulus from Chart: CS-2 at 155 °C: 199186720 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.000	570.00	2566.67	81.43	4.5029	0.0003625	36.11

$$MAEP = (4*B)/(3*(Do/t)) = (4*36.1)/(3*81.4) = 5.91 \text{ bars}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.433	570.00	2566.67	166.02	4.5029	0.0001293	12.88

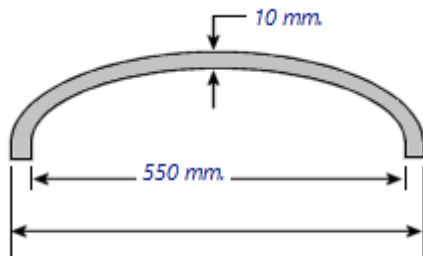
$$MAEP = (4*B)/(3*(Do/t)) = (4*12.9)/(3*166) = 1.03 \text{ bars}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.000	570.00	71808.20	81.43	50.0000	0.0001668	16.61

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.6)/(3*81.4) = 2.72 \text{ bars}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Top D end

Elliptical Head Design Information:					
Design Pressure	P	3.500 bars	Int. Design Temperature		155 °C
Material		SA-516 70	Ext. Design Temperature		155 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	550.000 mm.	Actual Stress		16.2 N./mm^2
Spec. Min. Thk t _e , or s _a		10.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		4473.1 cm^2	SG of Contents		0.000
Element Volume		33663.6 cm3	Weight of Contents		0.0 Kg
Empty Weight		43.4 Kg	Operating Weight		43.4 Kg
			Comp. All. Ope. Stress		133.0 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.875 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.438 mm.

Required Thickness due to Internal Pressure [t_r]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (3.5 \cdot 556 \cdot 0.99) / (2 \cdot 138 \cdot 0.85 - 0.2 \cdot 3.5)$$

$$= 0.8181 + 3.0000 = 3.8181 \text{ mm.}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm. per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 7) / (0.99 \cdot 556 + 0.2 \cdot 7)$$

$$= 29.9 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 10) / (1 \cdot 550 + 0.2 \cdot 10)$$

$$= 42.5 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [S_{act}]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (3.5 \cdot (0.99 \cdot 556 + 0.2 \cdot 7)) / (2 \cdot 0.85 \cdot 7)$$

= 16.164 N./mm²

Straight Flange Required Thickness:
 = $(P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 = $(3.5 \cdot 278) / (138 \cdot 0.85 - 0.6 \cdot 3.5) + 3$
 = 3.831 mm.

Straight Flange Maximum Allowable Working Pressure:
 = $(S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 = $(138 \cdot 0.85 \cdot 9) / (278 + 0.6 \cdot 9)$
 = 37.2 bars

Factor K, corroded condition [Kcor]:
 = $(2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 = $(2 + (556 / (2 \cdot 140))^2) / 6$
 = 0.985841

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 9.045 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 10, tr = 1.5, c = 3 mm., E* = 0.85
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.18$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 12, tr = 0.98, c = 3 mm., E* = 0.85
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.092$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.122 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	10.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.182	Temperature Difference	75 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.875 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.438 mm.

EXTERNAL PRESSURE RESULTS for: Top D end

Elliptical Head From 30 to 40 Ext. Chart: CS-2 at 155 °C

Elastic Modulus from Chart: CS-2 at 155 °C: 199186720 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
7.000	570.00	81.43	0.0017057	98.74

MAEP = $B / (K0 \cdot Do / t) = 98.7 / (0.9 \cdot 81.4) = 13.5$ bars



Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
1.479	570.00	385.49	0.0003603	35.88

$$MAEP = B / (K0 * Do / t) = 35.9 / (0.9 * 385) = 1.03 \text{ bars}$$

Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5.358 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	4.550 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	25.382 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	4.534 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	5.005 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	5.005 bars
App 27-4	= M.A.W.P.	4.122 bars

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 4.12 • 1
 = 5.358 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom D end	17.7	234.0	0.076	5.46
Shell	18.1	234.0	0.077	5.46
Top D end	17.7	234.0	0.076	5.46

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
B2_2"	Nozzle	138.00	138.00	1.000
H1_6"	Nozzle	138.00	138.00	1.000
L2_2"	Nozzle	138.00	138.00	1.000
L4_2"	Nozzle	138.00	138.00	1.000
L1_2"	Nozzle	138.00	138.00	1.000
L3_2"	Nozzle	138.00	138.00	1.000
A1_4"	Nozzle	138.00	138.00	1.000
A1_3"	Nozzle	118.00	118.00	1.000
B1_3"	Nozzle	138.00	138.00	1.000
S1_8"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom D end	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Top D end	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
B2_2"	1.16	225.00	0.005
H1_6"	7.26	225.00	0.032
L2_2"	1.36	225.00	0.006
L4_2"	1.36	225.00	0.006
L1_2"	1.36	225.00	0.006
L3_2"	1.36	225.00	0.006
A1_4"	3.80	225.00	0.017
A1_3"	2.28	216.00	0.011



B1_3"	3.80	225.00	0.017
S1_8"	5.88	225.00	0.026

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5.358 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	4.550 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	25.382 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	4.534 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	5.005 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	5.005 bars
App 27-4	= M.A.W.P.	4.122 bars

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 4.12 • 1
 = 5.358 bars

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom D end	26.2	234.0	0.112	5.68
Shell	26.8	234.0	0.115	5.66
Top D end	25.1	234.0	0.107	5.43

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
B2_2"	Nozzle	138.00	138.00	1.000
H1_6"	Nozzle	138.00	138.00	1.000
L2_2"	Nozzle	138.00	138.00	1.000
L4_2"	Nozzle	138.00	138.00	1.000
L1_2"	Nozzle	138.00	138.00	1.000
L3_2"	Nozzle	138.00	138.00	1.000
A1_4"	Nozzle	138.00	138.00	1.000
A1_3"	Nozzle	118.00	118.00	1.000
B1_3"	Nozzle	138.00	138.00	1.000
S1_8"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom D end	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Top D end	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
B2_2"	1.20	225.00	0.005
H1_6"	7.53	225.00	0.033
L2_2"	1.41	225.00	0.006
L4_2"	1.41	225.00	0.006
L1_2"	1.41	225.00	0.006
L3_2"	1.41	225.00	0.006
A1_4"	3.93	225.00	0.017
A1_3"	2.36	216.00	0.011

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HEXAGON

B1_3"	3.77	225.00	0.017
S1_8"	5.84	225.00	0.026

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	570	7	0.0017057	98.7411
20	30	2566.67	570	7	0.00036254	36.1087
30	40	No Calc	570	7	0.0017057	98.7411

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	10	4.47863	1.03419	13.5
20	30	10	6.4333	1.03419	5.91
30	40	10	4.47863	1.03419	13.5

Minimum

5.91

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	2566.67	71808.2	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	43.3685	33663.6	32.5264	34644.9	3.03579
20	30	323.841	564361	227.902	576741	22.6688
30	40	43.3685	33663.6	32.5264	34644.9	3.03579
Total		410	631687.75	292	646031.06	28

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset, Dtl. Cent. mm.	Description
10	Liqd	31.4966	...	-68.75	...	Liquid 10
10	Ins1	3.60136	...	-43.75	...	Ins: 10
10	Noz1	10.9634	...	-285.687	...	B2_2"
10	Wght	2	...	-137	...	Vortex breaker
20	Liqd	255.678	...	575	...	Liquid: 20
20	Ins1	26.892	...	1187.5	...	Ins: 20
20	Noz1	36.687	350.006	350	...	H1_6"
20	Noz1	10.9926	379.775	600	...	L2_2"
20	Noz1	8.34136	...	600	-379.775	L4_2"
20	Noz1	8.34136	...	1600	-379.775	L1_2"
20	Noz1	8.34136	379.775	1600	...	L3_2"
20	Noz1	38.5431	-379.775	1600	...	A1_4"
20	Noz1	19.215	-450.006	1600	...	A1_3"
20	Legs	64.6288	...	-565	...	LEGS
20	Wght	25	...	1250	...	Lifting WT
20	Wght	31	...	...	...	insu. CLIP WT
30	Ins1	3.60136	...	93.75	...	Ins: 30
30	Noz1	15.0065	200	200.511	...	B1_3"
30	Noz1	46.8904	-150	260.244	...	S1_8"

Total Weight of Each Detail Type:

Liquid	287.2
Insulation	34.1
Nozzles	203.3
Legs	64.6
Weights	58.0

Sum of the Detail Weights 647.2 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	439.3	439.3	439.3	439.3	439.3	439.3
Nozzles	203.3	203.3	203.3	203.3	203.3	203.3
Legs	64.6	64.6	64.6	64.6	64.6	64.6
Wld Weights	58.0	58.0	58.0	58.0	58.0	58.0
Insulation	...	...	34.1	34.1	34.1	34.1
Ope. Liquid	...	...	...	...	...	287.2
Test Liquid	...	631.3	...	...	...	...
Totals	765.3	1396.6	799.4	799.4	799.4	1086.5

Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the



weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	765.3 kg.
Shop Test Wt.	- Fabricated weight + water (full)	1396.6 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	799.4 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	799.4 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	799.4 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	1086.5 kg.
Field Test Wt.	- Empty weight + test medium (full)	1430.7 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	263.1 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	416.8 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	308.2 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm^2	Inside Surface Area cm^2
10	20	4473.11	4142.99
20	30	42529.3	41037.1
30	40	4473.11	4142.99

Total	51475.5	49323.0 cm2
Total	55.4	53.1 ft²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	62.969	94.4656	96.6121	...	62.969
20	Legs	35.3598	51.5079	70.9818	3.10141	35.3598
Legs	30	524.504	764.034	1052.9	...	524.504
30	40	111.903	111.903	145.546	4.03233	111.903

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	...	...	...
20	Legs	-62.969	-94.4656	-96.6121
Legs	30	636.406	875.936	1198.44
30	40	111.903	111.903	145.546

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	...	...	...
20	Legs	3.10141	3.10141	3.10141
Legs	30	4.03233	4.03233	4.03233
30	40	4.03233	4.03233	4.03233



Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
B2_2"	44.97	51.00	155	300	GR 1.1	63.334	4.122	44.970	4.122
H1_6"	15.60	19.60	155	150	GR 1.1	...	...	...	...
L2_2"	15.60	19.60	155	150	GR 1.1	...	...	...	...
L4_2"	15.60	19.60	155	150	GR 1.1	...	...	...	...
L1_2"	15.60	19.60	155	150	GR 1.1	...	...	...	...
L3_2"	15.60	19.60	155	150	GR 1.1	...	...	...	...
A1_4"	44.97	51.00	155	300	GR 1.1	44.781	22.674	44.970	22.674
A1_3"	89.85	102.00	155	600	GR 1.1	...	...	...	...
B1_3"	44.97	51.00	155	300	GR 1.1	48.759	18.696	44.970	18.696
S1_8"	44.97	51.00	155	300	GR 1.1	30.481	36.975	44.970	36.975
Min Rating	4.122	19.600, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg-f	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
B2_2"	77.851	0.50	53	326	4.122	4.122	Pass
A1_4"	138.176	0.50	214	653	4.122	22.674	Pass
B1_3"	111.125	0.50	120	489	4.122	18.696	Pass
S1_8"	248.539	0.50	856	1305	4.122	36.975	Pass

Evaluating Flange on Nozzle [B22]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(53) + 4 \cdot \text{abs}(326)77.851 \leq \pi \cdot 77.851^3 [(44.970 - 4.122) + .500 \cdot 44.970]$$

$$= 0.95738E+03 \leq 0.95738E+03 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle B22" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(44.970, 0.500 \cdot 44.970 + 44.970 - 957.4 / (\pi \cdot 77.851^3))$$

$$= \min(44.970, 4.122)$$

$$= 4.122 \text{ bars}$$

Evaluating Flange on Nozzle [A14]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(214) + 4 \cdot \text{abs}(653)138.176 \leq \pi \cdot 138.176^3 [(44.970 - 4.122) + .500 \cdot 44.970]$$

$$= 0.37848E+04 \leq 0.53529E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A14" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(44.970, 0.500 \cdot 44.970 + 44.970 - 3784.8 / (\pi \cdot 138.176^3))$$

$$= \min(44.970, 22.674)$$

$$= 22.674 \text{ bars}$$

Evaluating Flange on Nozzle [B13]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(120) + 4 \cdot \text{abs}(489)111.125 \leq \pi \cdot 111.125^3 [(44.970 - 4.122) + .500 \cdot 44.970]$$

$$= 0.21436E+04 \leq 0.27844E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle B13" [Pmax]:



$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(44.970, 0.500 \cdot 44.970 + 44.970 - 2143.6 / (\pi \cdot 111.125^3)) \\ &= \min(44.970, 18.696) \\ &= 18.696 \text{ bars} \end{aligned}$$

Evaluating Flange on Nozzle [S18"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(856) + 4 \cdot \text{abs}(1305) 248.539 \leq \pi \cdot 248.539^3 [(44.970 - 4.122) + .500 \cdot 44.970] \\ &= 0.14992E+05 \leq 0.31151E+05 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle S18" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(44.970, 0.500 \cdot 44.970 + 44.970 - 14992.1 / (\pi \cdot 248.539^3)) \\ &= \min(44.970, 36.975) \\ &= 36.975 \text{ bars} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 14.5432 Hz.

The Natural Frequency for the Vessel (Ope...) is 12.625 Hz.

The Natural Frequency for the Vessel (Filled) is 10.8506 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE-7 2010
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Importance Factor	1.0
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.19 \cdot 0.98) / (1 + 1.7 \cdot 3.4 \cdot 0.19)))$
 $= \min(0.85, 0.91)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	12.625 Hz
Natural Frequency of Vessel (Empty)	14.543 Hz
Natural Frequency of Vessel (Test)	10.851 Hz

Force Coefficient	[Cf] 0.555
Structure Height to Diameter ratio	4.272
Height to top of Structure	2652.500 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: D from Table 26.9.1
 $\alpha = 11.5 : Zg = 213360 \text{ mm.}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 129 + 300 = 429 \text{ mm.}$
 $= 1.41 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (1.41 ft.) $< 15 \text{ ft.}$
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/700)^{2/11.5}$
 $= 1.03$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 0.95, \text{ per Table 26.6-1}$

As there is No Hill Present: $[Kzt]$:
 $K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.03 \cdot 1 \cdot 0.95 \cdot 112^2)$$

$$= 31.3 \text{ psf } [153 \text{ Kgs/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 31.3 \cdot 0.85 \cdot 0.55 \cdot 2.41$$

$$= 35.7 \text{ lbs. } [16.2 \text{ Kgf}]$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m2
Bottom D end	428.7	0.000	0.000	0.000	1.030	1.000	153.042
Shell	1675.0	0.000	0.000	0.000	1.030	1.000	153.042
Top D end	2961.3	0.000	0.000	0.000	1.030	1.000	153.042

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm.
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm.
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure mm.
 Lc - Total length of conical section(s) of vessel, mm.
 tb - Uncorroded plate thickness at bottom of vessel, mm.
 V30 - Design Wind Speed provided by user, Km/hr
 Vc - Critical wind velocity, Km/hr
 Vw - Maximum wind speed at top of structure, Km/hr
 W - Total corroded weight of structure, Kgf
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kgf
 Z - Maximum amplitude of vibration at top of vessel, mm.
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, <= 0.32037E-06 (High); 0.32037E-06 < 0.40047E-06 (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0/2475 = 0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[2.07/8.12^2] \cdot 10^4 = 313$ [Geometry Violation]

Compute the vibration possibility. If Vp > 0.40047E-06 no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 961 / (2475 \cdot 610^2)$$

$$= 0.10431E-05$$

Since Vp is > 0.40047E-06 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf
Vortex breake	0.00	163.00	153.04	0.00



Lifting WT 0.00 1600.00 153.04 0.00
 insu. CLIP WT 0.00 350.00 153.04 0.00

The Natural Frequency for the Vessel (Ope...) is 12.625 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm^2	Wind Pressure Kgs/m^2	Element Wind Load Kgf
10	20	428.653	1184.4	2243.35	153.042	17.801
20	30	1675	1184.4	28129.5	153.042	223.207
30	40	2961.35	1184.4	2243.35	153.042	17.801

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-22	
Importance Factor		1.500
Force Modification Factor R		2.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	0.000
Force Factor		0.000
Consider Vertical Acceleration		No
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[OmegaO]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.33 = 0.22$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.14 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(12.60.75)$
 = 0.134 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = $(1/\text{Natural Frequency}) = (1/12.6)$
 = 0.079

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of $(1.7 \cdot 0.13, 0.079)$ per 12.8.2
 = 0.079

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 = $Sds / (R / I)$
 = $0.22 / (2/1.5)$
 = 0.163

Maximum value of C_s per equation 12.8-3 [Cs]:
 = $Sd1 / (T(R / I_e))$
 = $0.093 / (0.079 (2/1.5))$
 = 0.877

Minimum value of C_s per equation 15.4-1 [Cs]:
 = $\max(0.044 \cdot SDS \cdot I_e, 0.030)$
 = $\max(0.044 \cdot 0.22 \cdot 1.5, 0.030)$
 = 0.030

Total Base Shear [V]:
 = $C_s \cdot W$ (Equation 12.8-1):
 = $0.16 \cdot 1022$
 = 166.571 Kgf

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} \cdot V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x \cdot h_x^k / (\text{Sum of } W_i \cdot h_i^k)$ and k is an exponent which is related

to the period of Vibration.

In this case, the value of k was 1

The Natural Frequency for the Vessel (Ope...) is 12.625 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	25	94.4656	0.30504	0.19412	...
20	Legs	200	51.5079	1.33058	0.87205	...
Legs	30	1312.5	764.034	129.524	84.8886	...
30	40	2450	111.903	35.4116	33.8071	...

The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	248.847	...	...	...	...
20	Legs	75	17.801	0.30504	1.75961	0.030153
Legs	30	1112.5	258.809	166.571	268.52	226.215
30	40	2323.85	17.801	35.4116	1.75961	3.50041

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	...	2.62515	2.62515	0.0027497
20	Legs	17.801	2.62524	2.62548	0.002754
Legs	30	258.809	2.64457	2.67635	0.0027793
30	40	17.801	2.67709	2.67784	0.0027793

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	268.431	1677.69
20	30	268.431	1677.69
30	40	268.431	1677.69

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 12.375 Actual deflection : 2.678 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	175.929	123.81	2420567	1721489
20	30	175.929	123.81	2420567	1721489
30	40	175.929	123.81	2420567	1721489

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	165.597	280.795	-132.997	-134.774
20	Legs	165.597	280.795	-132.997	-134.774
Legs	30	165.597	280.795	-132.997	-134.774
30	40	165.597	280.795	-132.997	-134.774

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	6.8804	-2.13162	10.5333
20	30	6.8804	-2.13162	10.5333
30	40	6.8804	-2.13162	10.5333

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	...	...	...	...	...
20	Legs	0.049877	0.074825	0.076525	0.017667	0.017667
Legs	30	-0.50409	-0.4913	-0.50409	0.022971	0.022971
30	40	-0.088637	-0.088637	-0.088637	0.022971	0.022971

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	...	...	...	...	...
20	Legs	0.017667	0.010024	0.00017177	0.0033079	...
Legs	30	0.022971	1.52965	1.28866	0.50478	...
30	40	0.022971	0.010024	0.01994	0.0033079	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	...
20	Legs	...	...	...	0.00010931
Legs	30	...	...	...	0.98489
30	40	...	...	...	0.019037

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	...	...
20	Legs	...	...
Legs	30	...	...
30	40	...	...

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	...	...	...	...
20	Legs	...	...	...	...
Legs	30	...	...	...	...
30	40	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	Legs	...
Legs	30	...
30	40	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	133.00	0.0000	-0.0000
20	0.08	165.60		133.00	0.0005	
20	1.05	165.60	-2.06	133.00	0.0063	0.0155
30		165.60	-0.12	133.00		0.0009

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	133.00	0.0000	-0.0000
20	0.07	165.60		133.00	0.0004	
20	0.50	165.60	-1.51	133.00	0.0030	0.0114
30		165.60	-0.13	133.00		0.0010

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	133.00	0.0000	-0.0000
20	0.10	165.60		133.00	0.0006	
20	1.06	165.60	-2.04	133.00	0.0064	0.0154
30		165.60	-0.12	133.00		0.0009

Analysis of Load Case 4 : NP+OW+EQ+FS+BS



From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	133.00	0.0000	-0.0000
20	0.09	165.60		133.00	0.0006	
20	0.82	165.60	-1.80	133.00	0.0050	0.0136
30		165.60	-0.13	133.00		0.0010

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	280.79	0.00	134.77	0.0000	-0.0000
20	0.10	280.79		134.77	0.0003	
20	0.02	280.79	-1.03	134.77	0.0001	0.0077
30		280.79	-0.11	134.77		0.0009

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	280.79	0.00	134.77	0.0000	-0.0000
20	0.09	280.79		134.77	0.0003	
20		280.79	-0.53	134.77		0.0039
30		280.79	-0.11	134.77		0.0008

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	6.88	165.60		133.00	0.0415	
20	6.98	165.60		133.00	0.0422	
20	7.94	165.60		133.00	0.0480	
30	6.82	165.60		133.00	0.0412	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	6.88	165.60		133.00	0.0415	
20	6.97	165.60		133.00	0.0421	
20	7.70	165.60		133.00	0.0465	
30	6.83	165.60		133.00	0.0413	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-2.13	133.00		0.0160
20		165.60	-2.08	133.00		0.0157
20		165.60	-4.18	133.00		0.0314
30		165.60	-2.25	133.00		0.0169

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-2.13	133.00		0.0160
20		165.60	-2.07	133.00		0.0156
20		165.60	-3.93	133.00		0.0296
30		165.60	-2.26	133.00		0.0170

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	10.53	280.79		134.77	0.0375	
20	10.63	280.79		134.77	0.0379	
20	10.56	280.79		134.77	0.0376	
30	10.47	280.79		134.77	0.0373	



Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	10.53	280.79		134.77	0.0375	
20	10.63	280.79		134.77	0.0378	
20	10.05	280.79		134.77	0.0358	
30	10.47	280.79		134.77	0.0373	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	6.88	165.60		133.00	0.0415	
20	6.95	165.60		133.00	0.0420	
20	6.40	165.60		133.00	0.0386	
30	6.81	165.60		133.00	0.0412	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	6.88	165.60		133.00	0.0415	
20	6.93	165.60		133.00	0.0419	
20	6.53	165.60		133.00	0.0395	
30	6.82	165.60		133.00	0.0412	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	6.88	165.60		133.00	0.0415	
20	6.97	165.60		133.00	0.0421	
20	6.41	165.60		133.00	0.0387	
30	6.81	165.60		133.00	0.0412	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	6.88	165.60		133.00	0.0415	
20	6.95	165.60		133.00	0.0420	
20	6.40	165.60		133.00	0.0386	
30	6.81	165.60		133.00	0.0412	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	133.00	0.0000	-0.0000
20	0.09	165.60		133.00	0.0006	
20		165.60	-0.51	133.00		0.0039
30		165.60	-0.11	133.00		0.0008

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	6.88	138.00		110.83	0.0499	
20	6.97	138.00		110.83	0.0505	
20	6.41	138.00		110.83	0.0465	
30	6.81	138.00		110.83	0.0494	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		138.00	-2.13	110.83		0.0192
20		138.00	-2.07	110.83		0.0187



20	138.00	-2.65	110.83	0.0239
30	138.00	-2.24	110.83	0.0202

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	6.88	165.60		133.00	0.0415	
20	6.97	165.60		133.00	0.0421	
20	6.41	165.60		133.00	0.0387	
30	6.81	165.60		133.00	0.0412	

Absolute Maximum of the all of the Stress Ratio's 0.0505

Governing Element: Shell

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Bottom D end	6.9	...	...	...	6.9	+ Tens
Bottom D end	6.9	...	...	...	...	- Comp
Shell	6.9	0.1	...	...	7.0	+ Tens
Shell	6.9	0.1	...	...	...	- Comp
Shell	6.9	-0.5	...	...	6.4	+ Tens
Shell	6.9	-0.5	...	...	...	- Comp
Top D end	6.9	-0.1	...	...	6.8	+ Tens
Top D end	6.9	-0.1	...	...	...	- Comp



Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Liquid	548.911 mm.
Center of Gravity of the Insulation	1237.500 mm.
Center of Gravity of the Nozzles	1535.736 mm.
Center of Gravity of the Legs	-515.000 mm.
Center of Gravity of the Added Weights (Operating)	582.345 mm.
Center of Gravity of the Added Weights (Empty)	582.345 mm.
Center of Gravity of Bare Shell New and Cold	1237.500 mm.
Center of Gravity of Bare Shell Corroded	1237.500 mm.
Vessel CG in the Operating Condition	937.330 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	1124.131 mm.
Vessel CG in the Test Condition	1174.157 mm.

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RESULTS FOR LEGS : Operating Case Description: LEGS

Legs attached to: Shell

Section Properties : Single Angle L4X4X0.3125

USA AISC 1989 Steel Table

Overall Leg Length		1430.000	mm.
Effective Leg Length	Leglen	1200.000	mm.
Distance Leg Up Side of Vessel		150.000	mm.
Number of Legs	Nleg	3	
Cross Sectional Area for L4X4X0.3125	Aleg	15.484	cm ²
Section Inertia (strong axis)		154.422	cm ⁴
Section Inertia (weak axis)		154.422	cm ⁴
Section Modulus (strong axis)		21139.311	mm. ³
Section Modulus (weak axis)		21139.311	mm. ³
Radius of Gyration (strong axis)		31.496	mm.
Radius of Gyration (weak axis)		31.496	mm.

Leg Orientation - Diagonal

Overturning Moment at top of Legs		268.5	Kg-m.
Total Weight Load at top of Legs	W	1026.1	Kgf
Total Shear force at top of Legs		258.8	Kgf
Additional force in Leg due to Bracing	Fadd	0.0	Kgf
Occasional Load Factor	Occfac	1.333	
Effective Leg End Condition Factor	k	1.000	

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Pad Width along Circumference	C11P	150.000	mm.
Pad Length along Vessel Axis	C22P	150.000	mm.
Pad Thickness	Tpad	10.000	mm.

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (259) (247/372)$
 $= 171.69 \text{ Kg}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W/N_{\text{leg}} + (M_{\text{leg}} / (N_{\text{legm}} \cdot R_n))) / A_{\text{leg}}$
 $= (10063/3 + (2633073 / (1 \cdot 324))) / 1548$
 $= 7.41 \text{ N./mm}^2$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / N_{\text{leg}}) / A_{\text{leg}}$
 $= (1026/3) / 15.5$
 $= 2.17 \text{ N./mm}^2$

Computing Principal Axis and Inertias for Angle:

Leg lengths and thickness:	101.60	101.60	7.9375
Distance to geometric centroid:	28.448	28.448	
Arm about YY:	24.479	26.321	
Arm about ZZ:	22.352	24.479	
Leg areas:	8.0645	7.4345	
Geometric inertia components YY:	48.749	105.85	
Geometric inertia components ZZ:	109.66	44.940	
Geometric inertias Iy & Iz:	154.60	154.60	
Product of inertia:	92.027		
Mohrs Radius:	92.027		
Average Inertia:	154.60		

QFACT = 0.99670	FBZ	= 0.14844	
Principal Axis Inertias (Z&W)	= 62.576	246.63	
Angle to Principal Axis	= 45.000		



Distances to extreme fibers CW & CZ = 71.842 31.611
 FOB from Eq 5-5 = 1.2884
 Bending allowables Fby & Fbz = 0.16383 0.14844

Shear Center Coordinates Wo & Zo: 34.166 0.0000

Values for Elastic Flexural-Torsional Buckling Stress:

E, G, J, R0²: 29500. 11346. 0.78125E-01 4.9046
 AREA, LENGTH, Kw, Kz: 2.4000 47.244 1.0000 1.0000
 H, Few, Fez, Fej: 0.63110 322.05 81.713 75.305
 Fe computed from C4-1: 68.482

Initial (Kl/r)_{max}, & (Kl/r)_{equiv} = 59.692 65.204
 Final (Kl/r)_{max}, & Cc = 65.204 127.39
 Fa based on Eq 4-1 = 0.11673

Leg Bending and Axial Stresses vs. Allowables: (N./mm²)

	Actual	Allowable
Weak Axis Bending :	72.17	197.87
Strong Axis Bending :	41.62	218.38
Axial Compression :	7.41	155.60

UNITY CHECKS ARE: H1-1 0.000
 H1-2 0.000
 H1-3 0.603

AISC Unity Check : 0.603, should be <= to 1 for Code compliance, [Passed]

WRC 107 Stress Analysis for Leg to Shell Junction, Ope Condition

Rectangular Attachment Parameter C11 143.684 mm.
 Rectangular Attachment Parameter C22 130.950 mm.

Input Echo, WRC107/537 Item 1, Description: LEGS

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Cylindrical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	550.000	mm.
Vessel Thickness	Tv	10.000	mm.
Design Temperature	T1	155.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Rectangular
Parameter C11	C11	143.68 mm.
Parameter C22	C22	130.95 mm.
Thickness of Reinforcing Pad	Tpad	10.000 mm.
Pad Parameter C11P	C11p	150.000 mm.
Pad Parameter C22P	C22p	150.000 mm.
Design Internal Pressure	Dp	3.500 bars
Include Pressure Thrust		No
Vessel Centerline Direction Cosine	Vx	0.000
Vessel Centerline Direction Cosine	Vy	1.000
Vessel Centerline Direction Cosine	Vz	0.000
Nozzle Centerline Direction Cosine	Nx	1.000
Nozzle Centerline Direction Cosine	Ny	0.000
Nozzle Centerline Direction Cosine	Nz	0.000



Global Force (SUS)	Fx	0.0	Kgf
Global Force (SUS)	Fy	342.0	Kgf
Global Force (SUS)	Fz	0.0	Kgf
Global Moment (SUS)	Mx	0.0	Kg-m.
Global Moment (SUS)	My	0.0	Kg-m.
Global Moment (SUS)	Mz	13.5	Kg-m.

Internal Pressure (SUS)	P	3.50	bars
Include Pressure Thrust		No	

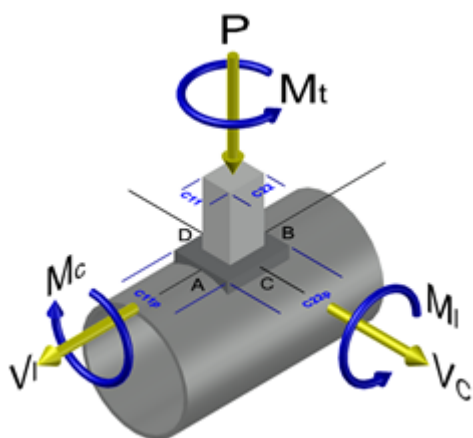
Global Force (OCC)	Fx	171.7	Kgf
Global Force (OCC)	Fy	827.7	Kgf
Global Force (OCC)	Fz	0.0	Kgf
Global Moment (OCC)	Mx	0.0	Kg-m.
Global Moment (OCC)	My	0.0	Kg-m.
Global Moment (OCC)	Mz	135.6	Kg-m.

Occasional Internal Pressure (OCC)	Pvar	0.00	bars
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Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSstained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	342.0	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-13.5	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 16.85

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.239	4C	2.617	(A,B)
N(PHI) / (P/Rm)	0.239	3C	1.946	(C,D)



M(PHI) / (P)	0.245	2C1	0.046	(A,B)
M(PHI) / (P)	0.245	1C	0.075	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.243	3A	0.691	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.275	1A	0.083	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.236	3B	1.867	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.245	1B	0.032	(A,B,C,D)
N(x) / (P/Rm)	0.245	3C	1.912	(A,B)
N(x) / (P/Rm)	0.245	4C	2.594	(C,D)
M(x) / (P)	0.241	1C1	0.079	(A,B)
M(x) / (P)	0.241	2C	0.047	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.243	4A	1.188	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.292	2A	0.040	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.236	4B	0.659	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.250	2B	0.051	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.7	0.7	-0.7	-0.7	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		1.3	-1.3	-1.3	1.3	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		1.9	-0.6	-1.9	0.6	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.2	0.2	-0.2	-0.2	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		2.0	-2.0	-2.0	2.0	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		2.2	-1.7	-2.2	1.7	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Str. Int.		2.2	1.7	2.2	1.7	1.5	1.5	1.5	1.5

Dimensionless Parameters used : Gamma = 40.21

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.266	4C	4.811	(A,B)
N(PHI) / (P/Rm)	0.266	3C	2.532	(C,D)
M(PHI) / (P)	0.266	2C1	0.020	(A,B)
M(PHI) / (P)	0.266	1C !	0.063	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.266	3A	1.521	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.266	1A	0.068	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.266	3B	3.308	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.266	1B	0.016	(A,B,C,D)
N(x) / (P/Rm)	0.266	3C	2.532	(A,B)
N(x) / (P/Rm)	0.266	4C	4.811	(C,D)
M(x) / (P)	0.266	1C1	0.046	(A,B)
M(x) / (P)	0.266	2C !	0.032	(C,D)



N(x) / (MC/(Rm**2 * Beta)) 0.266 4A 3.612 (A,B,C,D)
 M(x) / (MC/(Rm * Beta)) 0.266 2A 0.029 (A,B,C,D)
 N(x) / (ML/(Rm**2 * Beta)) 0.266 4B 1.572 (A,B,C,D)
 M(x) / (ML/(Rm * Beta)) 0.266 2B 0.025 (A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		3.0	3.0	-3.0	-3.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		3.4	-3.4	-3.4	3.4	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		6.3	-0.4	-6.3	0.4	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		1.4	1.4	-1.4	-1.4	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		5.4	-5.4	-5.4	5.4	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		6.8	-4.0	-6.8	4.0	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-1.6	-1.6	1.6	1.6
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-1.6	-1.6	1.6	1.6
Str. Int.		6.8	4.0	6.8	4.0	3.2	3.2	3.2	3.2

WRC 107 Stress Calculation for OCCasional loads:

Radial Load	P	171.7	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	827.7	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-135.6	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 16.85
 Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.9	-0.9	-0.9	-0.9	-0.7	-0.7	-0.7	-0.7
Circ. Bend. P		-1.6	1.6	-1.6	1.6	-2.6	2.6	-2.6	2.6
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		6.7	6.7	-6.7	-6.7	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		12.6	-12.6	-12.6	12.6	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		16.8	-5.2	-21.9	6.6	-3.3	1.9	-3.3	1.9
Long. Memb. P		-0.7	-0.7	-0.7	-0.7	-0.9	-0.9	-0.9	-0.9



Long. Bend. P	-2.8	2.8	-2.8	2.8	-1.7	1.7	-1.7	1.7
Long. Memb. MC	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC	-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML	2.4	2.4	-2.4	-2.4	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	19.8	-19.8	-19.8	19.8	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	18.8	-15.2	-25.6	19.5	-2.5	0.8	-2.5	0.8
Shear VC	-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL	0.0	0.0	0.0	0.0	-1.8	-1.8	1.8	1.8
Shear MT	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear	0.0	0.0	0.0	0.0	-1.8	-1.8	1.8	1.8
Str. Int.	18.8	15.2	25.6	19.5	4.8	3.8	4.8	3.8

Dimensionless Parameters used : Gamma = 40.21
 Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-4.1	-4.1	-4.1	-4.1	-2.2	-2.2	-2.2	-2.2
Circ. Bend. P		-4.2	4.2	-4.2	4.2	-12.9	12.9	-12.9	12.9
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		29.8	29.8	-29.8	-29.8	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		34.0	-34.0	-34.0	34.0	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		55.5	-4.2	-72.1	4.3	-15.1	10.7	-15.1	10.7
Long. Memb. P		-2.2	-2.2	-2.2	-2.2	-4.1	-4.1	-4.1	-4.1
Long. Bend. P		-9.4	9.4	-9.4	9.4	-6.6	6.6	-6.6	6.6
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		14.1	14.1	-14.1	-14.1	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		54.4	-54.4	-54.4	54.4	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		56.9	-33.0	-80.1	47.5	-10.8	2.5	-10.8	2.5
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-3.9	-3.9	3.9	3.9
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-3.9	-3.9	3.9	3.9
Str. Int.		56.9	33.0	80.1	47.5	17.3	12.3	17.3	12.3

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		5.6	5.9	5.6	5.9	5.6	5.9	5.6	5.9
Circ. Pm (OCC)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pm(TOTAL)		5.6	5.9	5.6	5.9	5.6	5.9	5.6	5.9
Circ. Pl (SUS)		0.7	0.7	-0.7	-0.7	0.0	0.0	-0.0	-0.0
Circ. Pl (OCC)		5.8	5.8	-7.6	-7.6	-0.7	-0.7	-0.7	-0.7
Circ. Pl(TOTAL)		6.5	6.5	-8.3	-8.3	-0.7	-0.7	-0.7	-0.7
Circ. Q (SUS)		1.3	-1.3	-1.3	1.3	0.0	0.0	-0.0	0.0
Circ. Q (OCC)		11.0	-11.0	-14.2	14.2	-2.6	2.6	-2.6	2.6
Circ. Q (TOTAL)		12.3	-12.3	-15.5	15.5	-2.6	2.6	-2.6	2.6

Long. Pm (SUS)	2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.8
Long. Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm(TOTAL)	2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.8
Long. Pl (SUS)	0.2	0.2	-0.2	-0.2	0.0	0.0	-0.0	-0.0
Long. Pl (OCC)	1.8	1.8	-3.1	-3.1	-0.9	-0.9	-0.9	-0.9
Long. Pl(TOTAL)	2.0	2.0	-3.3	-3.3	-0.9	-0.9	-0.9	-0.9
Long. Q (SUS)	2.0	-2.0	-2.0	2.0	0.0	0.0	-0.0	0.0
Long. Q (OCC)	17.0	-17.0	-22.5	22.5	-1.7	1.7	-1.7	1.7
Long. Q (TOTAL)	19.0	-19.0	-24.5	24.5	-1.7	1.7	-1.7	1.7
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear Pl (OCC)	0.0	0.0	0.0	0.0	-1.8	-1.8	1.8	1.8
Shear Pl(TOTAL)	0.0	0.0	0.0	0.0	-2.6	-2.6	2.6	2.6
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	5.6	5.9	5.6	5.9	5.6	5.9	5.6	5.9
Pm (SUS+OCC)	5.6	5.9	5.6	5.9	5.6	5.9	5.6	5.9
Pm+Pl (SUS)	6.2	6.6	4.9	5.2	5.7	6.1	5.7	6.1
Pm+Pl (SUS+OCC)	12.1	12.4	2.8	2.4	6.4	6.6	6.4	6.6
Pm+Pl+Q (Total)	24.3	14.3	25.0	24.0	5.5	9.1	5.5	9.1

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.90	138.00	Passed
Pm (SUS+OCC)	5.90	165.60	Passed
Pm+Pl (SUS)	6.57	207.00	Passed
Pm+Pl (SUS+OCC)	12.41	248.40	Passed
Pm+Pl+O (TOTAL)	25.04	462.42	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The P_m+P_l+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		13.7	14.1	13.7	14.1	13.7	14.1	13.7	14.1
Circ. Pm (OCC)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pm(TOTAL)		13.7	14.1	13.7	14.1	13.7	14.1	13.7	14.1
Circ. Pl (SUS)		3.0	3.0	-3.0	-3.0	0.0	0.0	-0.0	-0.0
Circ. Pl (OCC)		25.7	25.7	-33.9	-33.9	-2.2	-2.2	-2.2	-2.2
Circ. Pl(TOTAL)		28.6	28.6	-36.8	-36.8	-2.2	-2.2	-2.2	-2.2
Circ. Q (SUS)		3.4	-3.4	-3.4	3.4	0.0	0.0	-0.0	0.0
Circ. Q (OCC)		29.8	-29.8	-38.2	38.2	-12.9	12.9	-12.9	12.9
Circ. Q (TOTAL)		33.2	-33.2	-41.6	41.6	-12.9	12.9	-12.9	12.9
Long. Pm (SUS)		6.9	6.9	6.9	6.9	6.9	6.9	6.9	6.9
Long. Pm (OCC)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm(TOTAL)		6.9	6.9	6.9	6.9	6.9	6.9	6.9	6.9



Long. P1 (SUS)	1.4	1.4	-1.4	-1.4	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)	12.0	12.0	-16.3	-16.3	-4.1	-4.1	-4.1	-4.1
Long. P1(TOTAL)	13.4	13.4	-17.7	-17.7	-4.1	-4.1	-4.1	-4.1
Long. Q (SUS)	5.4	-5.4	-5.4	5.4	0.0	0.0	-0.0	0.0
Long. Q (OCC)	44.9	-44.9	-63.8	63.8	-6.6	6.6	-6.6	6.6
Long. Q (TOTAL)	50.3	-50.3	-69.2	69.2	-6.6	6.6	-6.6	6.6
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-1.6	-1.6	1.6	1.6
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-3.9	-3.9	3.9	3.9
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-5.5	-5.5	5.5	5.5
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	13.7	14.1	13.7	14.1	13.7	14.1	13.7	14.1
Pm (SUS+OCC)	13.7	14.1	13.7	14.1	13.7	14.1	13.7	14.1
Pm+P1 (SUS)	16.7	17.0	10.8	11.1	14.1	14.4	14.1	14.4
Pm+P1 (SUS+OCC)	42.3	42.7	23.1	22.8	14.2	14.5	14.2	14.5
Pm+P1+Q (Total)	75.6	39.6	80.1	58.4	11.2	26.6	11.2	26.6

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.08	138.00	Passed
Pm (SUS+OCC)	14.08	165.60	Passed
Pm+P1 (SUS)	17.04	207.00	Passed
Pm+P1 (SUS+OCC)	42.70	248.40	Passed
Pm+P1+Q (TOTAL)	80.07	462.42	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

Bolting Size Requirement for Leg Baseplates :

Baseplate Material	SA-516 70
Baseplate Allowable Stress	SBA 137.90 N./mm2
Baseplate Length	B 154.0000 mm.
Baseplate Width	D 154.0000 mm.
Baseplate Thickness	BTHK 16.0000 mm.
Leg Dimension Along Baseplate Length	d 101.6000 mm.
Leg Dimension Along Baseplate Width	b 101.6000 mm.
Bolt Circle Diameter	BCD 520.0000 mm.
Bolt Material	SA-36
Bolt Allowable Tensile Stress	STBA 149.90 N./mm2
Bolt Allowable Shear Stress	SBShear 99.93 N./mm2
Anchor Bolt Nominal Diameter	BOD 20.0000 mm.
Number of Anchor Bolts in Tension per Leg	NB 0
Total Number of Anchors Bolt per Leg	NBT 1
Bolt Corrosion Allowance	bca 1.5000 mm.
Ultimate 28-day Concrete Strength	FCPRIME 35.000 N./mm2

Leg Baseplate and Bolting Analysis, including Moments

Shear Stress in a Single Bolt [rb]:
 = Shear Force/(Bolt Area • Number of Bolts)
 = 259/(1.46 • 3)
 = 5.8 N./mm². Must be less than 99.9 N./mm².

Plate thickness required (5/8"; 16mm) per D. Moss



Since there is only 1 bolt, the standard computation for this geometry is not possible. A reasonable bolt size should be selected. Per Megyesy, use a 1 inch (25.4 mm) bolt.

Review notes about nozzle loadings and supports in the Warnings report.

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RESULTS FOR LEGS : HydroTest Case Description: LEGS

Legs attached to: Shell

Section Properties : Single Angle L4X4X0.3125

USA AISC 1989 Steel Table

Overall Leg Length		1430.000	mm.
Effective Leg Length	Leglen	1200.000	mm.
Distance Leg Up Side of Vessel		150.000	mm.
Number of Legs	Nleg	3	
Cross Sectional Area for L4X4X0.3125	Aleg	15.484	cm ²
Section Inertia (strong axis)		154.422	cm ⁴
Section Inertia (weak axis)		154.422	cm ⁴
Section Modulus (strong axis)		21139.311	mm. ³
Section Modulus (weak axis)		21139.311	mm. ³
Radius of Gyration (strong axis)		31.496	mm.
Radius of Gyration (weak axis)		31.496	mm.

Leg Orientation - Diagonal

Overturning Moment at top of Legs		88.6	Kg-m.
Total Weight Load at top of Legs	W	1336.2	Kgf
Total Shear force at top of Legs		85.4	Kgf
Additional force in Leg due to Bracing	Fadd	0.0	Kgf
Occasional Load Factor	Occfac	1.333	
Effective Leg End Condition Factor	k	1.000	

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Pad Width along Circumference	C11P	150.000	mm.
Pad Length along Vessel Axis	C22P	150.000	mm.
Pad Thickness	Tpad	10.000	mm.

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (85.4) (247/372)$
 $= 56.66 \text{ Kg}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W/N_{\text{leg}} + (M_{\text{leg}} / (N_{\text{legm}} \cdot R_n))) / A_{\text{leg}}$
 $= (13103/3 + (868914 / (1 \cdot 324))) / 1548$
 $= 4.55 \text{ N./mm}^2$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / N_{\text{leg}}) / A_{\text{leg}}$
 $= (1336/3) / 15.5$
 $= 2.82 \text{ N./mm}^2$

Computing Principal Axis and Inertias for Angle:

Leg lengths and thickness:	101.60	101.60	7.9375
Distance to geometric centroid:	28.448	28.448	
Arm about YY:	24.479	26.321	
Arm about ZZ:	22.352	24.479	
Leg areas:	8.0645	7.4345	
Geometric inertia components YY:	48.749	105.85	
Geometric inertia components ZZ:	109.66	44.940	
Geometric inertias Iy & Iz:	154.60	154.60	
Product of inertia:	92.027		
Mohrs Radius:	92.027		
Average Inertia:	154.60		

QFACT = 0.99670	FBZ	= 0.14844	
Principal Axis Inertias (Z&W)	= 62.576	246.63	
Angle to Principal Axis	= 45.000		



Distances to extreme fibers CW & CZ = 71.842 31.611
 FOB from Eq 5-5 = 1.2884
 Bending allowables Fby & Fbz = 0.16383 0.14844

Shear Center Coordinates Wo & Zo: 34.166 0.0000

Values for Elastic Flexural-Torsional Buckling Stress:

E, G, J, R0²: 29500. 11346. 0.78125E-01 4.9046
 AREA, LENGTH, Kw, Kz: 2.4000 47.244 1.0000 1.0000
 H, Few, Fez, Fej: 0.63110 322.05 81.713 75.305
 Fe computed from C4-1: 68.482

Initial (Kl/r)max, & (Kl/r)equiv = 59.692 65.204
 Final (Kl/r)max, & Cc = 65.204 127.39
 Fa based on Eq 4-1 = 0.11673

Leg Bending and Axial Stresses vs. Allowables: (N./mm²)

	Actual	Allowable
Weak Axis Bending :	23.82	197.87
Strong Axis Bending :	13.73	218.38
Axial Compression :	4.55	155.60

UNITY CHECKS ARE: H1-1 0.000
 H1-2 0.000
 H1-3 0.212

AISC Unity Check : 0.212, should be <= to 1 for Code compliance, [Passed]

WRC 107 Stress Analysis for Leg to Shell Junction, Test Condition

Rectangular Attachment Parameter C11 143.684 mm.
 Rectangular Attachment Parameter C22 130.950 mm.

Input Echo, WRC107/537 Item 1, Description: LEGS

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Cylindrical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	550.000	mm.
Vessel Thickness	Tv	10.000	mm.
Design Temperature	T1	155.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	234.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	234.00	N./mm ²
Attachment Type	Type	Rectangular	
Parameter C11	C11	143.68	mm.
Parameter C22	C22	130.95	mm.
Thickness of Reinforcing Pad	Tpad	10.000	mm.
Pad Parameter C11P	C11p	150.000	mm.
Pad Parameter C22P	C22p	150.000	mm.
Design Internal Pressure	Dp	3.500	bars
Include Pressure Thrust		No	
Vessel Centerline Direction Cosine	Vx	0.000	
Vessel Centerline Direction Cosine	Vy	1.000	
Vessel Centerline Direction Cosine	Vz	0.000	
Nozzle Centerline Direction Cosine	Nx	1.000	
Nozzle Centerline Direction Cosine	Ny	0.000	
Nozzle Centerline Direction Cosine	Nz	0.000	
Global Force (SUS)	Fx	0.0	Kgf
Global Force (SUS)	Fy	445.4	Kgf

Global Force (SUS)	Fz	0.0	Kgf
Global Moment (SUS)	Mx	0.0	Kg-m.
Global Moment (SUS)	My	0.0	Kg-m.
Global Moment (SUS)	Mz	17.6	Kg-m.

Internal Pressure (SUS)	P	3.50	bars
Include Pressure Thrust		No	

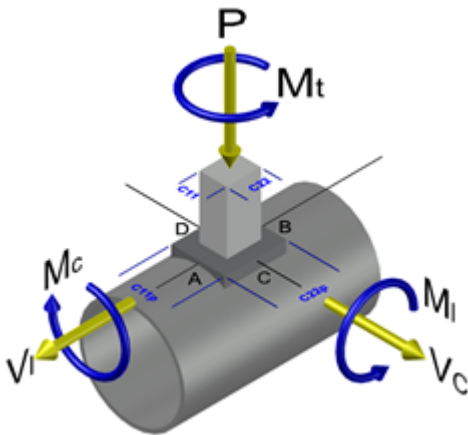
Global Force (OCC)	Fx	56.7	Kgf
Global Force (OCC)	Fy	273.1	Kgf
Global Force (OCC)	Fz	0.0	Kgf
Global Moment (OCC)	Mx	0.0	Kg-m.
Global Moment (OCC)	My	0.0	Kg-m.
Global Moment (OCC)	Mz	44.8	Kg-m.

Occasional Internal Pressure (OCC)	Pvar	2.14	bars
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Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	445.4	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-17.6	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 16.85

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.239	4C	2.617	(A,B)
N(PHI) / (P/Rm)	0.239	3C	1.946	(C,D)
M(PHI) / (P)	0.245	2C1	0.046	(A,B)
M(PHI) / (P)	0.245	1C	0.075	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.243	3A	0.691	(A,B,C,D)



$M(\text{PHI}) / (MC/(Rm * \text{Beta})) = 0.275$ 1A 0.083 (A,B,C,D)
 $N(\text{PHI}) / (ML/(Rm^{**2} * \text{Beta})) = 0.236$ 3B 1.867 (A,B,C,D)
 $M(\text{PHI}) / (ML/(Rm * \text{Beta})) = 0.245$ 1B 0.032 (A,B,C,D)

 $N(x) / (P/Rm) = 0.245$ 3C 1.912 (A,B)
 $N(x) / (P/Rm) = 0.245$ 4C 2.594 (C,D)
 $M(x) / (P) = 0.241$ 1C1 0.079 (A,B)
 $M(x) / (P) = 0.241$ 2C 0.047 (C,D)
 $N(x) / (MC/(Rm^{**2} * \text{Beta})) = 0.243$ 4A 1.188 (A,B,C,D)
 $M(x) / (MC/(Rm * \text{Beta})) = 0.292$ 2A 0.040 (A,B,C,D)
 $N(x) / (ML/(Rm^{**2} * \text{Beta})) = 0.236$ 4B 0.659 (A,B,C,D)
 $M(x) / (ML/(Rm * \text{Beta})) = 0.250$ 2B 0.051 (A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.9	0.9	-0.9	-0.9	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		1.6	-1.6	-1.6	1.6	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		2.5	-0.8	-2.5	0.8	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.3	0.3	-0.3	-0.3	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		2.6	-2.6	-2.6	2.6	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		2.9	-2.2	-2.9	2.2	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Str. Int.		2.9	2.2	2.9	2.2	2.0	2.0	2.0	2.0

Dimensionless Parameters used : $\Gamma = 40.21$

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
$N(\text{PHI}) / (P/Rm)$	0.266	4C	4.811	(A,B)
$N(\text{PHI}) / (P/Rm)$	0.266	3C	2.532	(C,D)
$M(\text{PHI}) / (P)$	0.266	2C1	0.020	(A,B)
$M(\text{PHI}) / (P)$	0.266	1C !	0.063	(C,D)
$N(\text{PHI}) / (MC/(Rm^{**2} * \text{Beta}))$	0.266	3A	1.521	(A,B,C,D)
$M(\text{PHI}) / (MC/(Rm * \text{Beta}))$	0.266	1A	0.068	(A,B,C,D)
$N(\text{PHI}) / (ML/(Rm^{**2} * \text{Beta}))$	0.266	3B	3.308	(A,B,C,D)
$M(\text{PHI}) / (ML/(Rm * \text{Beta}))$	0.266	1B	0.016	(A,B,C,D)
$N(x) / (P/Rm)$	0.266	3C	2.532	(A,B)
$N(x) / (P/Rm)$	0.266	4C	4.811	(C,D)
$M(x) / (P)$	0.266	1C1	0.046	(A,B)
$M(x) / (P)$	0.266	2C !	0.032	(C,D)
$N(x) / (MC/(Rm^{**2} * \text{Beta}))$	0.266	4A	3.612	(A,B,C,D)
$M(x) / (MC/(Rm * \text{Beta}))$	0.266	2A	0.029	(A,B,C,D)
$N(x) / (ML/(Rm^{**2} * \text{Beta}))$	0.266	4B	1.572	(A,B,C,D)



M(x) / (ML/(Rm * Beta)) 0.266 2B 0.025 (A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		3.9	3.9	-3.9	-3.9	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		4.4	-4.4	-4.4	4.4	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		8.3	-0.5	-8.3	0.5	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		1.8	1.8	-1.8	-1.8	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		7.0	-7.0	-7.0	7.0	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		8.9	-5.2	-8.9	5.2	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-2.1	-2.1	2.1	2.1
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-2.1	-2.1	2.1	2.1
Str. Int.		8.9	5.2	8.9	5.2	4.2	4.2	4.2	4.2

WRC 107 Stress Calculation for OCCasional loads:

Radial Load	P	56.7	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	273.1	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-44.8	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 16.85

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.3	-0.3	-0.3	-0.3	-0.2	-0.2	-0.2	-0.2
Circ. Bend. P		-0.5	0.5	-0.5	0.5	-0.9	0.9	-0.9	0.9
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		2.2	2.2	-2.2	-2.2	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		4.2	-4.2	-4.2	4.2	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		5.6	-1.7	-7.2	2.2	-1.1	0.6	-1.1	0.6
Long. Memb. P		-0.2	-0.2	-0.2	-0.2	-0.3	-0.3	-0.3	-0.3
Long. Bend. P		-0.9	0.9	-0.9	0.9	-0.5	0.5	-0.5	0.5
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-1.4	-1.4	-1.4	-1.4	-0.7	-0.7	-0.7	-0.7
Circ. Bend. P		-1.4	1.4	-1.4	1.4	-4.3	4.3	-4.3	4.3
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		9.8	9.8	-9.8	-9.8	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		11.2	-11.2	-11.2	11.2	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		18.3	-1.4	-23.8	1.4	-5.0	3.5	-5.0	3.5
Long. Memb. P		-0.7	-0.7	-0.7	-0.7	-1.4	-1.4	-1.4	-1.4
Long. Bend. P		-3.1	3.1	-3.1	3.1	-2.2	2.2	-2.2	2.2
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		4.7	4.7	-4.7	-4.7	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		17.9	-17.9	-17.9	17.9	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		18.8	-10.9	-26.4	15.7	-3.5	0.8	-3.5	0.8
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-1.3	-1.3	1.3	1.3
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-1.3	-1.3	1.3	1.3
Str. Int.		18.8	10.9	26.4	15.7	5.7	4.1	5.7	4.1

Vessel Stress Summation at Attachment Junction (N./mm2)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		5.6	5.9	5.6	5.9	5.6	5.9	5.6	5.9
Circ. Pm (OCC)		3.4	3.6	3.4	3.6	3.4	3.6	3.4	3.6
Circ. Pm(TOTAL)		9.0	9.5	9.0	9.5	9.0	9.5	9.0	9.5
Circ. Pl (SUS)		0.9	0.9	-0.9	-0.9	0.0	0.0	-0.0	-0.0
Circ. Pl (OCC)		1.9	1.9	-2.5	-2.5	-0.2	-0.2	-0.2	-0.2
Circ. Pl(TOTAL)		2.8	2.8	-3.4	-3.4	-0.2	-0.2	-0.2	-0.2
Circ. Q (SUS)		1.6	-1.6	-1.6	1.6	0.0	0.0	-0.0	0.0
Circ. Q (OCC)		3.6	-3.6	-4.7	4.7	-0.9	0.9	-0.9	0.9
Circ. Q (TOTAL)		5.3	-5.3	-6.3	6.3	-0.9	0.9	-0.9	0.9
Long. Pm (SUS)		2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.8
Long. Pm (OCC)		1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7



Long. Pm(TOTAL)	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5
Long. P1 (SUS)	0.3	0.3	-0.3	-0.3	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)	0.6	0.6	-1.0	-1.0	-0.3	-0.3	-0.3	-0.3
Long. P1(TOTAL)	0.9	0.9	-1.3	-1.3	-0.3	-0.3	-0.3	-0.3
Long. Q (SUS)	2.6	-2.6	-2.6	2.6	0.0	0.0	-0.0	0.0
Long. Q (OCC)	5.6	-5.6	-7.4	7.4	-0.5	0.5	-0.5	0.5
Long. Q (TOTAL)	8.2	-8.2	-10.0	10.0	-0.5	0.5	-0.5	0.5
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-1.6	-1.6	1.6	1.6
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	5.6	5.9	5.6	5.9	5.6	5.9	5.6	5.9
Pm (SUS+OCC)	9.0	9.5	9.0	9.5	9.0	9.5	9.0	9.5
Pm+P1 (SUS)	6.4	6.8	4.7	5.0	5.9	6.2	5.9	6.2
Pm+P1 (SUS+OCC)	11.8	12.3	5.6	6.1	9.2	9.7	9.2	9.7
Pm+P1+Q (Total)	17.0	9.9	6.8	13.1	8.4	10.6	8.4	10.6

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.90	234.00	Passed
Pm (SUS+OCC)	9.52	280.79	Passed
Pm+P1 (SUS)	6.78	350.99	Passed
Pm+P1 (SUS+OCC)	12.32	421.19	Passed
Pm+P1+Q (TOTAL)	17.02	701.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		13.7	14.1	13.7	14.1	13.7	14.1	13.7	14.1
Circ. Pm (OCC)		8.4	8.6	8.4	8.6	8.4	8.6	8.4	8.6
Circ. Pm(TOTAL)		22.1	22.7	22.1	22.7	22.1	22.7	22.1	22.7
Circ. P1 (SUS)		3.9	3.9	-3.9	-3.9	0.0	0.0	-0.0	-0.0
Circ. P1 (OCC)		8.5	8.5	-11.2	-11.2	-0.7	-0.7	-0.7	-0.7
Circ. P1(TOTAL)		12.3	12.3	-15.0	-15.0	-0.7	-0.7	-0.7	-0.7
Circ. Q (SUS)		4.4	-4.4	-4.4	4.4	0.0	0.0	-0.0	0.0
Circ. Q (OCC)		9.8	-9.8	-12.6	12.6	-4.3	4.3	-4.3	4.3
Circ. Q (TOTAL)		14.2	-14.2	-17.0	17.0	-4.3	4.3	-4.3	4.3
Long. Pm (SUS)		6.9	6.9	6.9	6.9	6.9	6.9	6.9	6.9
Long. Pm (OCC)		4.2	4.2	4.2	4.2	4.2	4.2	4.2	4.2
Long. Pm(TOTAL)		11.1	11.1	11.1	11.1	11.1	11.1	11.1	11.1
Long. P1 (SUS)		1.8	1.8	-1.8	-1.8	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)		4.0	4.0	-5.4	-5.4	-1.4	-1.4	-1.4	-1.4
Long. P1(TOTAL)		5.8	5.8	-7.2	-7.2	-1.4	-1.4	-1.4	-1.4



Long. Q (SUS)	7.0	-7.0	-7.0	7.0	0.0	0.0	-0.0	0.0
Long. Q (OCC)	14.8	-14.8	-21.1	21.1	-2.2	2.2	-2.2	2.2
Long. Q (TOTAL)	21.9	-21.9	-28.1	28.1	-2.2	2.2	-2.2	2.2
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-2.1	-2.1	2.1	2.1
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-1.3	-1.3	1.3	1.3
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-3.4	-3.4	3.4	3.4
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	13.7	14.1	13.7	14.1	13.7	14.1	13.7	14.1
Pm (SUS+OCC)	22.1	22.7	22.1	22.7	22.1	22.7	22.1	22.7
Pm+P1 (SUS)	17.6	17.9	9.9	10.2	14.3	14.6	14.3	14.6
Pm+P1 (SUS+OCC)	34.5	35.0	7.1	7.7	22.3	22.8	22.3	22.8
Pm+P1+Q (Total)	48.7	25.8	24.2	32.0	18.2	27.0	18.2	27.0

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.08	234.00	Passed
Pm (SUS+OCC)	22.70	280.79	Passed
Pm+P1 (SUS)	17.93	350.99	Passed
Pm+P1 (SUS+OCC)	35.02	421.19	Passed
Pm+P1+Q (TOTAL)	48.70	701.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

Bolting Size Requirement for Leg Baseplates :

Baseplate Material		SA-516 70
Baseplate Allowable Stress	SBA	137.90 N./mm2
Baseplate Length	B	154.0000 mm.
Baseplate Width	D	154.0000 mm.
Baseplate Thickness	BTHK	16.0000 mm.
Leg Dimension Along Baseplate Length	d	101.6000 mm.
Leg Dimension Along Baseplate Width	b	101.6000 mm.
Bolt Circle Diameter	BCD	520.0000 mm.
Bolt Material		SA-36
Bolt Allowable Tensile Stress	STBA	149.90 N./mm2
Bolt Allowable Shear Stress	SBShear	99.93 N./mm2
Anchor Bolt Nominal Diameter	BOD	20.0000 mm.
Number of Anchor Bolts in Tension per Leg	NB	0
Total Number of Anchors Bolt per Leg	NBT	1
Bolt Corrosion Allowance	bca	1.5000 mm.
Ultimate 28-day Concrete Strength	FCPRIME	35.000 N./mm2

Leg Baseplate and Bolting Analysis, including Moments

Shear Stress in a Single Bolt [rb]:
 = Shear Force/(Bolt Area • Number of Bolts)
 = 259/(1.46 • 3)
 = 5.8 N./mm². Must be less than 99.9 N./mm².

Plate thickness required (5/8"; 16mm) per D. Moss

Since there is only 1 bolt, the standard computation for this geometry is not possible. A reasonable bolt size should be



selected. Per Megyesy, use a 1 inch (25.4 mm) bolt.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: B22"

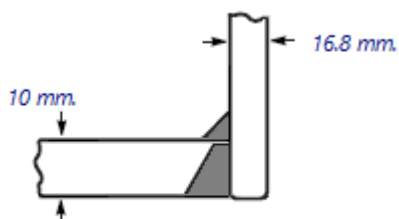
From: 10

Pressure for Reinforcement Calculations	P	3.623	bars
Temperature for Internal Pressure	Temp	155	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	155	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	550.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	10.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: B22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.62 \cdot 556 \cdot 0.89 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 3.62)$
 $= 0.6503$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.62 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 3.62)$
 $= 0.0747$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3930 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: B22".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3930	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 3.7197	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 4.5000	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(4.5, 4.5)]$
 $= 4.5000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.39, 4.5)$
 $= 4.5000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	10.5,	Allowable	:	138.0 N./mm2	Passed
Expansion	:	0.0,	Allowable	:	334.5 N./mm2	Passed
Occasional	:	0.3,	Allowable	:	183.5 N./mm2	Passed
Shear	:	10.1,	Allowable	:	96.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)



Govrn. thk, tg = 10, tr = 0.65, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.093, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 3.62/51.00 = 0.071

Weld Size Calculations, Description: B22"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.7703 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 218.1454 mm.

Input Echo, WRC107/537 Item 1, Description: B22" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	990.000	mm.
Vessel Thickness	Tv	10.000	mm.
Design Temperature	T1	155.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	50.800	mm.
Nozzle Thickness	Tn	16.764	mm.
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²
Design Internal Pressure	Dp	3.623	bars
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-326.3	Kgf
Longitudinal Shear (SUS)	(V1) V1	288.4	Kgf
Circumferential Shear (SUS)	(Vc) V2	288.4	Kgf
Circumferential Moment (SUS)	(Mc) M1	37.8	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	37.8	Kg-m.

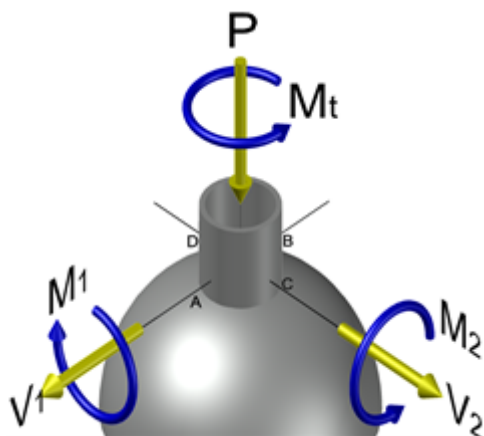


Torsional Moment (SUS) Mt 48.9 Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 84.328 + 2 \cdot 1.65 \cdot \sqrt{(501.5 (10.0 - 3.0))}$$

$$= 279.851 \text{ mm.}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	-326.3	Kgf
Circumferential Shear	(VC) V2	288.4	Kgf
Longitudinal Shear	(VL) V1	288.4	Kgf
Circumferential Moment	(MC) M1	37.8	Kg-m.
Longitudinal Moment	(ML) M2	37.8	Kg-m.
Torsional Moment	MT	48.9	Kg-m.

Dimensionless Param: U = 0.71 TAU = 5.00 (2.56) RHO = 0.51

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.05611	(A,B,C,D)
$M(x) / P$	SP 1	0.06393	(A,B,C,D)
$N(x) * T * \text{SQRT}(Rm * T) / MC$	SM 1	0.07445	(A,B,C,D)
$M(x) * \text{SQRT}(Rm * T) / MC$	SM 1	0.13833	(A,B,C,D)
$N(x) * T * \text{SQRT}(Rm * T) / ML$	SM 1	0.07445	(A,B,C,D)
$M(x) * \text{SQRT}(Rm * T) / ML$	SM 1	0.13833	(A,B,C,D)
$N(y) * T / P$	SP 1	0.09295	(A,B,C,D)
$M(y) / P$	SP 1	0.01943	(A,B,C,D)
$N(y) * T * \text{SQRT}(Rm * T) / MC$	SM 1	0.08918	(A,B,C,D)
$M(y) * \text{SQRT}(Rm * T) / MC$	SM 1	0.04864	(A,B,C,D)
$N(y) * T * \text{SQRT}(Rm * T) / ML$	SM 1	0.08918	(A,B,C,D)
$M(y) * \text{SQRT}(Rm * T) / ML$	SM 1	0.04864	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00



Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		3.7	3.7	3.7	3.7	3.7	3.7	3.7	3.7
Rad. Bend. P		25.1	-25.1	25.1	-25.1	25.1	-25.1	25.1	-25.1
Rad. Memb. MC		0.0	0.0	0.0	0.0	-9.5	-9.5	9.5	9.5
Rad. Bend. MC		0.0	0.0	0.0	0.0	-106.0	106.0	106.0	-106.0
Rad. Memb. ML		-9.5	-9.5	9.5	9.5	0.0	0.0	0.0	0.0
Rad. Bend. ML		-106.0	106.0	106.0	-106.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-86.8	75.1	144.3	-117.9	-86.8	75.1	144.3	-117.9
Tang. Memb. P		6.1	6.1	6.1	6.1	6.1	6.1	6.1	6.1
Tang. Bend. P		7.6	-7.6	7.6	-7.6	7.6	-7.6	7.6	-7.6
Tang. Memb. MC		0.0	0.0	0.0	0.0	-11.4	-11.4	11.4	11.4
Tang. Bend. MC		0.0	0.0	0.0	0.0	-37.3	37.3	37.3	-37.3
Tang. Memb. ML		-11.4	-11.4	11.4	11.4	0.0	0.0	0.0	0.0
Tang. Bend. ML		-37.3	37.3	37.3	-37.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-35.0	24.4	62.4	-27.4	-35.0	24.4	62.4	-27.4
Shear VC		3.1	3.1	-3.1	-3.1	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-3.1	-3.1	3.1	3.1
Shear MT		6.1	6.1	6.1	6.1	6.1	6.1	6.1	6.1
Tot. Shear		9.2	9.2	3.1	3.1	3.1	3.1	9.2	9.2
Str. Int.		88.4	76.7	144.4	118.0	87.0	75.3	145.3	118.8

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		12.9	12.9	12.9	12.9	12.9	12.9	12.9	12.9
Rad. Pl (SUS)		-5.8	-5.8	13.2	13.2	-5.8	-5.8	13.2	13.2
Rad. Q (SUS)		-81.0	81.0	131.1	-131.1	-81.0	81.0	131.1	-131.1
Long. Pm (SUS)		12.9	12.9	12.9	12.9	12.9	12.9	12.9	12.9
Long. Pl (SUS)		-5.3	-5.3	17.5	17.5	-5.3	-5.3	17.5	17.5
Long. Q (SUS)		-29.7	29.7	44.9	-44.9	-29.7	29.7	44.9	-44.9
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		3.1	3.1	-3.1	-3.1	-3.1	-3.1	3.1	3.1
Shear Q (SUS)		6.1	6.1	6.1	6.1	6.1	6.1	6.1	6.1
Pm (SUS)		12.9	12.9	12.9	12.9	12.9	12.9	12.9	12.9
Pm+Pl (SUS)		10.4	10.4	31.9	31.9	10.4	10.4	31.9	31.9
Pm+Pl+Q (Total)		75.5	89.6	157.3	105.1	74.1	88.2	158.2	105.9

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.89	138.00	Passed
Pm+Pl (SUS)	31.94	207.00	Passed
Pm+Pl+Q (TOTAL)	158.16	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.





Input, Nozzle Desc: H16"

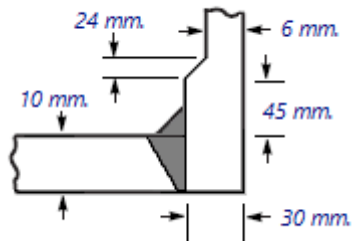
From: 20

Pressure for Reinforcement Calculations	P	3.573	bars
Temperature for Internal Pressure	Temp	155	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	155	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	550.00	mm.
Design Length of Section	L	2566.6665	mm.
Shell Finished (Minimum) Thickness	t	10.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		400.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	6.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	45.0000	mm.
Height of Beveled Transition	L'	24.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: H16"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	6.000 in.
Actual Thickness Used in Calculation	0.236 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.57 \cdot 278 / (138 \cdot 1 - 0.6 \cdot 3.57)$
 $= 0.7210$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.57 \cdot 79.2 / (138 \cdot 1 - 0.6 \cdot 3.57)$
 $= 0.2054$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.5073 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 24 / 0.5773$
 $= 41.5692$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

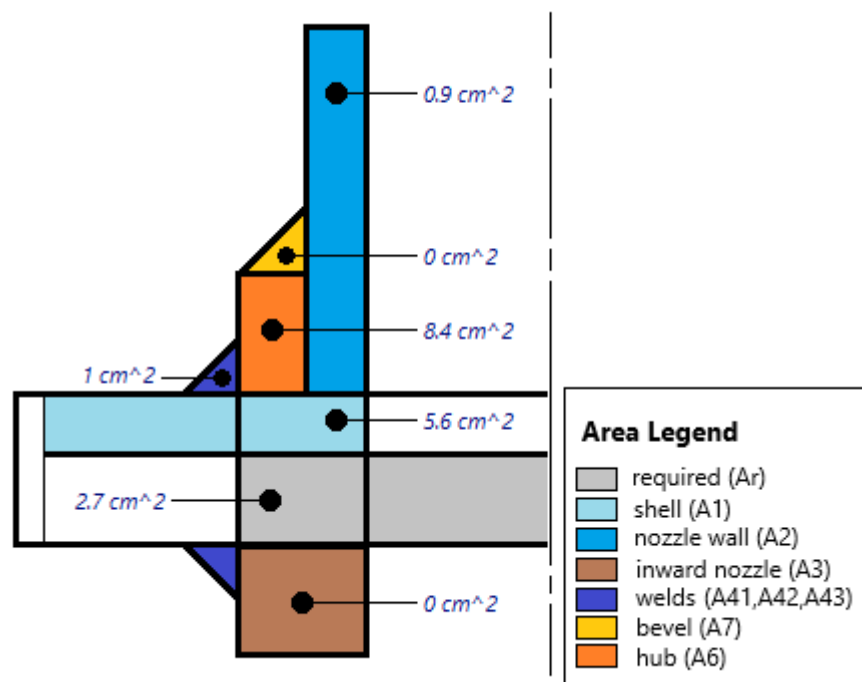
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	316.8000 mm.
Parallel to Vessel Wall, opening length	d	158.4000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1.142	2.719	NA
Area in Shell	A1	9.946	5.650	NA
Area in Nozzle Wall	A2	0.978	0.872	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	8.400	8.400	NA
TOTAL AREA AVAILABLE	Atot	20.324	15.922	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F (1 - fr1))$ per UG-37(d)
 $= 0.5 (158 \cdot 3.43 \cdot 1 + 2 \cdot 27 \cdot 3.43 \cdot 1 (1 - 1))$
 $= 2.719 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 158 (1 \cdot 7 - 1 \cdot 3.43) - 2 \cdot 27 (1 \cdot 7 - 1 \cdot 3.43) \cdot (1 - 1)$
 $= 5.650 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 17.5) (3 - 0.51) 1$
 $= 0.872 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(17.5, 150, 45)) \cdot (30 - 6) \cdot 1$$

$$= 8.400 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 3.5073 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 3.5073 \text{ mm}.$$

Available Nozzle Neck Thickness = 6.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 6$, $t_r = 0.21$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.068$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	-29 $^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 10$, $t_r = 0.72$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.1$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	-29 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.57/19.60 = 0.182$

Weld Size Calculations, Description: H16"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 7.0000 \text{ mm}$.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (2.72 - 5.65 + 2 \cdot 27 \cdot 1 \cdot (1 \cdot 7 - 3.43)) \cdot 138)$$

$$= \max(0, -1413.42) \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - C_{an} / 707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (9.27 + 1 - 0 \cdot 1) \cdot 138$$

$$= 14454.94 \text{ Kg}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv \\
 &= (9.27 + 0 + 1 + (3.78)) \cdot 138 \\
 &= 19774.00 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S \\
 &= (9.27 + 0 + 1 + 0 + (3.78)) \cdot 138 \\
 &= 19774.00 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\
 &= (3.14/2.0) \cdot 212 \cdot 10 \cdot 0.49 \cdot 138 \\
 &= 23005. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\
 &= (3.14 \cdot 92.7) \cdot (30 - 3) \cdot 0.7 \cdot 138 \\
 &= 77452. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\
 &= (3.14/2.0) \cdot 212 \cdot (10 - 3) \cdot 0.74 \cdot 138 \\
 &= 24319. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (23005 + 77452) = 100457 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (23005 + 0 + 24319 + 0) = 47324 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (23005 + 24319 + 0) = 47324 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 100456	Kgf, must exceed W = 0 Kgf	or W1 = 14454 Kgf
Path 2-2 = 47323	Kgf, must exceed W = 0 Kgf	or W2 = 19773 Kgf
Path 3-3 = 47323	Kgf, must exceed W = 0 Kgf	or W3 = 19773 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 29 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 21.3338 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 181.3338 mm.

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Input, Nozzle Desc: L22"

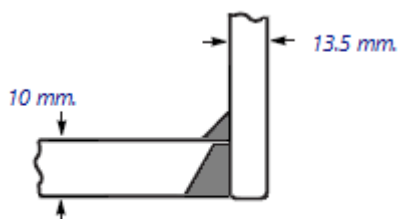
From: 20

Pressure for Reinforcement Calculations	P	3.550	bars
Temperature for Internal Pressure	Temp	155	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	155	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	550.00	mm.
Design Length of Section	L	2566.6665	mm.
Shell Finished (Minimum) Thickness	t	10.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		650.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	209.5500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: L22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.55 \cdot 278 / (138 \cdot 1 - 0.6 \cdot 3.55)$
 $= 0.7164$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.55 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 3.55)$
 $= 0.0732$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3780 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: L22".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3780	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 3.7164	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 4.5000	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(4.5, 4.5)]$
 $= 4.5000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.38, 4.5)$
 $= 4.5000$ mm.

Available Nozzle Neck Thickness = 13.4620 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 10, tr = 0.72, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.1$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $3.55/19.60 = 0.181$

Weld Size Calculations, Description: L22"

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.

[Skipping the nozzle attachment weld strength calculations. Per UW-15\(b\)\(2\) the nozzles exempted by UG-36\(c\)\(3\)\(a\) \(small nozzles\) do not require a weld strength check.](#)

The Drop for this Nozzle is : 2.7598 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.3098 mm.

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Input, Nozzle Desc: L42"

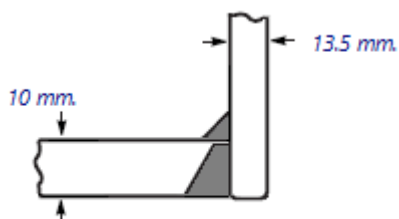
From: 20

Pressure for Reinforcement Calculations	P	3.550	bars
Temperature for Internal Pressure	Temp	155	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	155	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	550.00	mm.
Design Length of Section	L	2566.6665	mm.
Shell Finished (Minimum) Thickness	t	10.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		650.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	209.5500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: L42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.55 \cdot 278 / (138 \cdot 1 - 0.6 \cdot 3.55)$
 $= 0.7164$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.55 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 3.55)$
 $= 0.0732$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3780 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: L42".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3780	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 3.7164	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 4.5000	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(4.5, 4.5)]$
 $= 4.5000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.38, 4.5)$
 $= 4.5000$ mm.

Available Nozzle Neck Thickness = 13.4620 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 10, tr = 0.72, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.1$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $3.55/19.60 = 0.181$

Weld Size Calculations, Description: L42"

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 2.7598 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.3098 mm.

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Input, Nozzle Desc: L12"

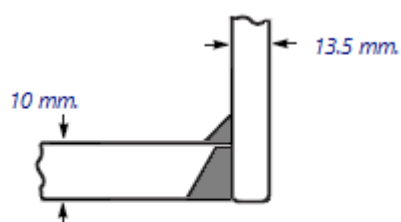
From: 20

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	155	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	155	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	550.00	mm.
Design Length of Section	L	2566.6665	mm.
Shell Finished (Minimum) Thickness	t	10.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1650.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	209.5500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: L12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 278 / (138 \cdot 1 - 0.6 \cdot 3.5)$
 $= 0.7062$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 3.5)$
 $= 0.0721$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3780 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: L12".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3780	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 3.7062	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 4.5000	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(4.5, 4.5)]$
 $= 4.5000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(3.38, 4.5)$
 $= 4.5000$ mm.

Available Nozzle Neck Thickness = 13.4620 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 10, tr = 0.71, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.1$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $3.50/19.60 = 0.179$

Weld Size Calculations, Description: L12"

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 2.7598 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.3098 mm.

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Input, Nozzle Desc: L32"

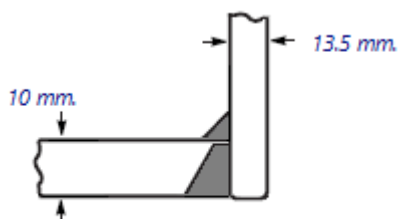
From: 20

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	155	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	155	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	550.00	mm.
Design Length of Section	L	2566.6665	mm.
Shell Finished (Minimum) Thickness	t	10.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1650.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	209.5500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: L32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 278 / (138 \cdot 1 - 0.6 \cdot 3.5)$
 $= 0.7062$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 3.5)$
 $= 0.0721$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3780 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: L32".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3780	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 3.7062	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 4.5000	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(4.5, 4.5)]$
 $= 4.5000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.38, 4.5)$
 $= 4.5000$ mm.

Available Nozzle Neck Thickness = 13.4620 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 10, tr = 0.71, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.1$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $3.50/19.60 = 0.179$

Weld Size Calculations, Description: L32"

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 2.7598 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.3098 mm.

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Input, Nozzle Desc: A14"

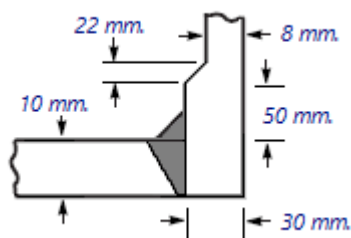
From: 20

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	155	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	155	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	550.00	mm.
Design Length of Section	L	2566.6665	mm.
Shell Finished (Minimum) Thickness	t	10.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1650.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	50.0000	mm.
Height of Beveled Transition	L'	22.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	209.5500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: A14"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	4.000 in.
Actual Thickness Used in Calculation	0.315 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 278 / (138 \cdot 1 - 0.6 \cdot 3.5)$
 $= 0.7062$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 53.8 / (138 \cdot 1 - 0.6 \cdot 3.5)$
 $= 0.1367$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4806 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 22 / 0.5773$
 $= 38.1051$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

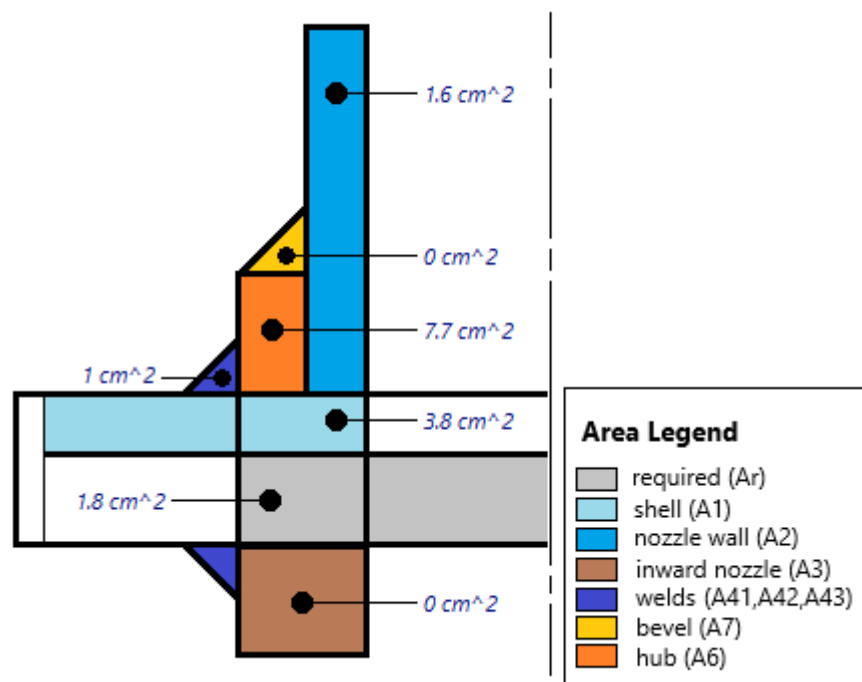
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	215.2000	mm.
Parallel to Vessel Wall, opening length	d	107.6000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	0.760	1.847	NA
Area in Shell	A1	6.772	3.838	NA
Area in Nozzle Wall	A2	1.702	1.582	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	7.700	7.700	NA
TOTAL AREA AVAILABLE	Atot	17.174	14.120	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F (1 - fr1))$ per UG-37(d)
 $= 0.5 (108 \cdot 3.43 \cdot 1 + 2 \cdot 27 \cdot 3.43 \cdot 1 (1 - 1))$
 $= 1.847 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 108 (1 \cdot 7 - 1 \cdot 3.43) - 2 \cdot 27 (1 \cdot 7 - 1 \cdot 3.43) \cdot (1 - 1)$
 $= 3.838 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 17.5) (5 - 0.48) 1$
 $= 1.582 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2}$
 $= (2 \cdot \min(17.5, 210, 50)) \cdot (30 - 8) \cdot 1$
 $= 7.700 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.4806 \text{ mm.}$
 Wall Thickness per UG16(b), $tr_{16b} = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb_1 = 3.7062 \text{ mm.}$
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 4.5000 \text{ mm.}$
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb_3 = 8.7300 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[8.73, \max(4.5, 4.5)]$
 $= 4.5000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(3.48, 4.5)$
 $= 4.5000 \text{ mm.}$

Available Nozzle Neck Thickness = 8.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	42.1,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	302.9 N./mm ²	Passed
Occasional	:	1.8,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	30.6,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 8$, $tr = 0.14$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.027$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 10$, $tr = 0.71$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.1$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.50/51.00 = 0.069$

Weld Size Calculations, Description: A14"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 7.0000 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.9000 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f r_1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (1.85 - 3.84 + 2 \cdot 27 \cdot 1 \cdot (1 \cdot 7 - 3.43)) \cdot 138)$
 $= \max(0, -90.95) \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f r_2) \cdot S_v$
 $= (9.28 + 1 - 0 \cdot 1) \cdot 138$
 $= 14468.11 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S_v$
 $= (9.28 + 0 + 1 + (3.78)) \cdot 138$
 $= 19787.16 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$
 $= (9.28 + 0 + 1 + 0 + (3.78)) \cdot 138$
 $= 19787.16 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 162 \cdot 10 \cdot 0.49 \cdot 138$
 $= 17502. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 67.3) \cdot (30 - 3) \cdot 0.7 \cdot 138$
 $= 56230. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 162 \cdot (10 - 3) \cdot 0.74 \cdot 138$
 $= 18503. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (17502 + 56230) = 73733 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (17502 + 0 + 18503 + 0) = 36005 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (17502 + 18503 + 0) = 36005 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 73732	Kgf, must exceed W = 0 Kgf	or W1 = 14468 Kgf
Path 2-2 = 36005	Kgf, must exceed W = 0 Kgf	or W2 = 19787 Kgf
Path 3-3 = 36005	Kgf, must exceed W = 0 Kgf	or W3 = 19787 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24.6 bars



Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 12.1381 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 231.6881 mm.

Input Echo, WRC297 Item 1, Description: A14" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	550.000	mm.
Shell Thickness	10.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	101.600	mm.
Nozzle Thickness	30.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	3.50	bars
Radial Load	P	-652.66	Kgf
Circumferential Shear	Vc	489.50	Kgf
Longitudinal Shear	Vl	652.66	Kgf
Circumferential Moment	Mc	130.48	Kg-m.
Longitudinal Moment	Ml	169.63	Kg-m.
Torsional Moment	Mt	195.72	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

**Warning - The ratio, Dn/Tn (5.387) is < 10
 check the limitations of WRC bulletin 297.**

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 161.6 + 2 \cdot 1.65 \cdot \sqrt{(281.5(10.0 - 3.0))}$$

$$= 308.088 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	563.000 mm.
Nozzle Outside Diameter	(d) =	161.600 mm.
Vessel Thickness used	(T) =	7.000 mm.
Nozzle Thickness used	(t) =	27.000 mm.
T / t	=	0.259
d / t	=	10.000



$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.574$$

Nr/P	=	0.062
Mr/P	=	0.120
M0/P	=	0.036
N0/P	=	0.072
MrD/Mc	=	0.276
NrDL/Mc	=	0.093
M0d/Mc	=	0.083
N0DL/Mc	=	0.084
MrD/M1	=	0.095
NrDL/M1	=	0.070
M0D/M1	=	0.028
N0DL/M1	=	0.092

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	28	-28	28	-28
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-19	-19	19	19
Outplane Bending (ML)	-34	34	34	-34
Normal Pressure Stress	13	14	13	14
Outplane Stress Summary	-2	10	105	-20

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	93	-93	93	-93
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-14	-14	14	14
Inplane Bending (ML)	-119	119	119	-119
Inplane Pressure Stress	6	6	6	6
Inplane Stress Summary	-25	25	242	-183

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	6	6	6	6
Shear Stress Summary	9	9	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	29	30	242	183

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	28	-28	28	-28



Outplane Membrane (Mc)	-13	-13	13	13
Outplane Bending (Mc)	-80	80	80	-80
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	6	6	6	6

Outplane Stress Summary	-49	55	138	-79
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	93	-93	93	-93
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-267	267	267	-267
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	13	14	13	14

Inplane Stress Summary	-167	181	398	-324
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-3	-3	3	3
Torsional Shear (Mt)	6	6	6	6

Shear Stress Summary	3	3	10	10
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	167	181	398	324
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-19	-19	19	19
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	1	0	1

Hoop Stress Summary	-9	-8	29	29
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-4	-4	4	4
Axial Bending (ML)	-5	5	5	-5
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-4	-3	15	-3
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	2	2	1	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	10	9	29	33

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-13	-13	13	13
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	1	0	1
Hoop Stress Summary	-3	-2	23	24

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-14	14	14	-14
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-13	7	24	-14

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	2	2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	13	10	26	39

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	13	14	13	14	13	14	13	14
Circ. Pl (SUS)	-10	-10	28	28	-7	-7	23	23
Circ. Q (SUS)	-6	6	62	-62	-174	174	360	-360
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-6	-6	22	22	-4	-4	22	22
Long. Q (SUS)	-26	26	212	-212	-52	52	108	-108
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-3	-3	3	3
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	13.0	14.0	13.0	14.0	13.0	14.0	13.0	14.0
Pm+Pl (SUS)	5.0	5.7	41.3	42.3	7.6	8.4	37.0	37.9
Pm+Pl+Q (Total)	28.5	29.3	240.1	184.1	168.1	181.1	396.3	323.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	138.00	Passed
Pm+Pl (SUS)	42.28	207.00	Passed
Pm+Pl+Q (TOTAL)	396.31	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	0	1	0	1	0	1	0	1
Circ. Pl (SUS)	-10	-10	28	28	-4	-4	22	22
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-4	-4	4	4	-3	-3	3	3
Long. Q (SUS)	-1	1	9	-9	-10	10	18	-18
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0
Pm+Pl (SUS)	10.0	9.0	28.0	29.0	4.0	3.0	22.0	23.0
Pm+Pl+Q (Total)	10.2	9.2	28.1	34.1	13.1	10.2	22.6	38.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1.00	138.00	Passed
Pm+Pl (SUS)	29.00	207.00	Passed
Pm+Pl+Q (TOTAL)	38.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: A13"

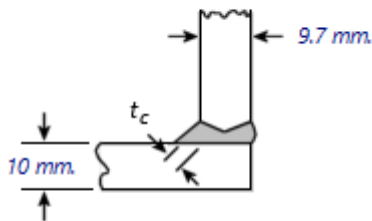
From: 20

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	155	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	155	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	550.00	mm.
Design Length of Section	L	2566.6665	mm.
Shell Finished (Minimum) Thickness	t	10.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1650.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm.
Inside Projection	h	100.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
ASME Code Weld Type per UW-16		b	
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



UW-16 (b)

Reinforcement CALCULATION, Description: A13"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, t_r [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 278 / (138 \cdot 1 - 0.6 \cdot 3.5)$
 $= 0.7062$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, t_{rn} [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.5 \cdot 44.5 / (118 \cdot 1 + 0.4 \cdot 3.5)$
 $= 0.1317$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.5044 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	150.8618 mm.
Parallel to Vessel Wall, opening length	d	75.4309 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	16.8364 mm.
Normal to Vessel Wall, Inward		9.3364 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: A13".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.5044$ mm.
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000$ mm.
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 3.7062$ mm.
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 4.5000$ mm.
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000$ mm.
Wall Thickness per table UG-45	$tb_3 = 7.8000$ mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[7.8, \max(4.5, 4.5)]$
 $= 4.5000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(t_a, t_b)$
 $= \max(3.5, 4.5)$
 $= 4.5000$ mm.



Available Nozzle Neck Thickness = 9.7345 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.73, tr = 0.13, c = 3 mm., E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.02$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.73, tr = 0.13, c = 3 mm., E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.02$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.50/102.00 = 0.034$

Weld Size Calculations, Description: A13"

Intermediate Calc. for nozzle/shell Welds Tmin 6.7345 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.7142 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.

The Drop for this Nozzle is : 3.6162 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 460.0000 mm.

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Input, Nozzle Desc: B13"

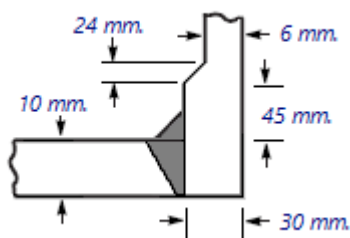
From: 30

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	155	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	155	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	550.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	10.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	200.0000	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	6.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	45.0000	mm.
Height of Beveled Transition	L'	24.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	122.2760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: B13"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.236 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 556 \cdot 0.99 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 3.5)$
 $= 0.6953$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 41.1 / (138 \cdot 1 - 0.6 \cdot 3.5)$
 $= 0.1044$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3244 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 24 / 0.5773$
 $= 41.5692$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

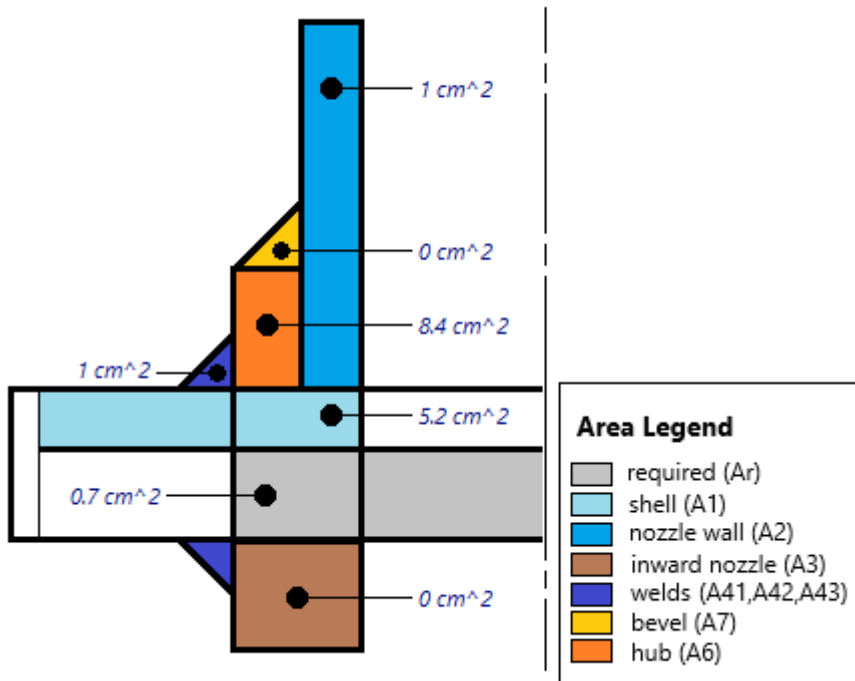
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	188.3926	mm.
Parallel to Vessel Wall, opening length	d	94.1963	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	0.655	0.696	NA
Area in Shell	A1	5.939	5.201	NA
Area in Nozzle Wall	A2	1.069	0.988	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	8.400	8.400	NA
TOTAL AREA AVAILABLE	Atot	16.408	15.589	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 60.77 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(94.2 \cdot 1.48 \cdot 1 + 2 \cdot 27 \cdot 1.48 \cdot 1(1 - 1))$
 $= 0.696 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 94.2 (1 \cdot 7 - 1 \cdot 1.48) - 2 \cdot 27$
 $(1 \cdot 7 - 1 \cdot 1.48) \cdot (1 - 1)$
 $= 5.201 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$
 $= (2 \cdot 17.5) (3 - 0.32) / \sin(71.4)$
 $= 0.988 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$

= 1.000 cm²

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(17.5, 122, 45)) \cdot (30 - 6) \cdot 1$$

$$= 8.400 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 6, tr = 0.1, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.035, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 10, tr = 0.7, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.099, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 3.50/51.00 = 0.069

Weld Size Calculations, Description: B13"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (0.7 - 5.2 + 2 \cdot 27 \cdot 1 \cdot (1 \cdot 7 - 1.48)) \cdot 138)$$

$$= \max(0, -2143.07) \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - Can / .707)^2 \cdot fr_2) \cdot Sv$$

$$= (9.39 + 1 - 0 \cdot 1) \cdot 138$$

$$= 14617.35 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot Sv$$

$$= (9.39 + 0 + 1 + (3.78)) \cdot 138$$

$$= 19936.40 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (9.39 + 0 + 1 + 0 + (3.78)) \cdot 138$$

= 19936.40 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 156 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 16904. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 62.6) \cdot (30 - 3) \cdot 0.7 \cdot 138$$

$$= 52277. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 156 \cdot (10 - 3) \cdot 0.74 \cdot 138$$

$$= 17870. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (16904 + 52277) = 69181 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (16904 + 0 + 17870 + 0) = 34775 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (16904 + 17870 + 0) = 34775 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 69181	Kgf, must exceed W = 0 Kgf	or W1 = 14617 Kgf
Path 2-2 = 34774	Kgf, must exceed W = 0 Kgf	or W2 = 19936 Kgf
Path 3-3 = 34774	Kgf, must exceed W = 0 Kgf	or W3 = 19936 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 15.1 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

Note : Checking Nozzle in the Latitudinal direction

Reinforcement CALCULATION, Description: B13"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.236 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 3.5 \cdot 556 \cdot 0.99 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 3.5)$$

$$= 0.6953 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 3.5 \cdot 41.1 / (138 \cdot 1 - 0.6 \cdot 3.5)$$

$$= 0.1044 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3244 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 = (Hub Thickness - Neck Thickness) / tan(30)
 = 24/0.5773
 = 41.5692 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	164.4000	mm.
Parallel to Vessel Wall, opening length	d	82.2000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: B13".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3244	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 3.6953	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 4.5000	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[7.8, max(4.5, 4.5)]
 = 4.5000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.32, 4.5)
 = 4.5000 mm.

Available Nozzle Neck Thickness = 6.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	67.7,	Allowable	:	138.0 N./mm2	Passed
Expansion	:	0.0,	Allowable	:	277.3 N./mm2	Passed
Occasional	:	2.3,	Allowable	:	183.5 N./mm2	Passed
Shear	:	49.4,	Allowable	:	96.6 N./mm2	Passed

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of
 the expansion stress allowable.*

Weld Size Calculations, Description: B13"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 35.9337 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 169.0884 mm.

Warning:



The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: B13" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	990.000	mm.
Vessel Thickness	Tv	10.000	mm.
Design Temperature	T1	155.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	76.200	mm.
Nozzle Thickness	Tn	30.000	mm.
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²
Design Internal Pressure	Dp	3.500	bars
Include Pressure Thrust		No	

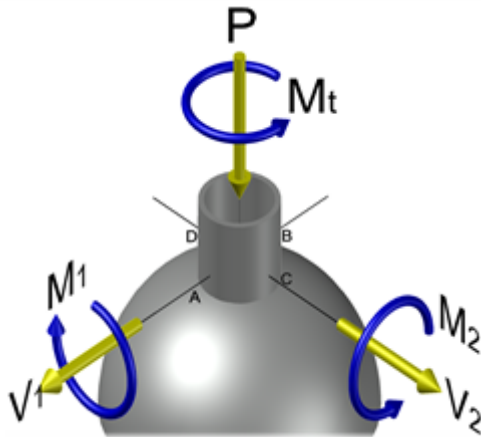
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-489.5	Kgf
Longitudinal Shear (SUS)	(V1) V1	432.7	Kgf
Circumferential Shear (SUS)	(Vc) V2	432.7	Kgf
Circumferential Moment (SUS)	(Mc) M1	85.1	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	85.1	Kg-m.
Torsional Moment (SUS)	Mt	110.1	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 136.2 + 2 \cdot 1.65 \cdot \sqrt{(501.5 (10.0 - 3.0))}$
 $= 331.723 \text{ mm.}$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	-489.5	Kgf
Circumferential Shear	(VC) V2	432.7	Kgf
Longitudinal Shear	(VL) V1	432.7	Kgf
Circumferential Moment	(MC) M1	85.1	Kg-m.
Longitudinal Moment	(ML) M2	85.1	Kg-m.
Torsional Moment	MT	110.1	Kg-m.

Dimensionless Param: $U = 1.15 \text{ TAU} = 5.00 (2.02) \text{ RHO} = 0.26$

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.03502	(A,B,C,D)
$M(x) / P$	SP 1	0.03755	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.03710	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.06432	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.03710	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.06432	(A,B,C,D)
$N(y) * T / P$	SP 1	0.05183	(A,B,C,D)
$M(y) / P$	SP 1	0.01013	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.05061	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01812	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.05061	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01812	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	3.4	3.4	3.4	3.4	3.4	3.4	3.4	3.4
Rad.	Bend. P	22.1	-22.1	22.1	-22.1	22.1	-22.1	22.1	-22.1
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-10.7	-10.7	10.7	10.7
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-111.0	111.0	111.0	-111.0
Rad.	Memb. ML	-10.7	-10.7	10.7	10.7	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-111.0	111.0	111.0	-111.0	0.0	0.0	0.0	0.0
Tot. Rad.	Str.	-96.1	81.7	147.1	-118.9	-96.1	81.7	147.1	-118.9



Tang. Memb. P	5.1	5.1	5.1	5.1	5.1	5.1	5.1	5.1
Tang. Bend. P	6.0	-6.0	6.0	-6.0	6.0	-6.0	6.0	-6.0
Tang. Memb. MC	0.0	0.0	0.0	0.0	-14.6	-14.6	14.6	14.6
Tang. Bend. MC	0.0	0.0	0.0	0.0	-31.3	31.3	31.3	-31.3
Tang. Memb. ML	-14.6	-14.6	14.6	14.6	0.0	0.0	0.0	0.0
Tang. Bend. ML	-31.3	31.3	31.3	-31.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-34.8	15.8	56.8	-17.6	-34.8	15.8	56.8	-17.6
Shear VC	2.8	2.8	-2.8	-2.8	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-2.8	-2.8	2.8	2.8
Shear MT	5.3	5.3	5.3	5.3	5.3	5.3	5.3	5.3
Tot. Shear	8.1	8.1	2.5	2.5	2.5	2.5	8.1	8.1
Str. Int.	97.2	82.6	147.2	119.0	96.2	81.7	147.9	119.6

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5
Rad.	Pl (SUS)	-7.2	-7.2	14.1	14.1	-7.2	-7.2	14.1	14.1
Rad.	Q (SUS)	-88.9	88.9	133.0	-133.0	-88.9	88.9	133.0	-133.0
Long.	Pm (SUS)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5
Long.	Pl (SUS)	-9.5	-9.5	19.6	19.6	-9.5	-9.5	19.6	19.6
Long.	Q (SUS)	-25.3	25.3	37.2	-37.2	-25.3	25.3	37.2	-37.2
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	2.8	2.8	-2.8	-2.8	-2.8	-2.8	2.8	2.8
Shear	Q (SUS)	5.3	5.3	5.3	5.3	5.3	5.3	5.3	5.3
Pm (SUS)		12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5
Pm+Pl (SUS)		7.1	7.1	33.3	33.3	7.1	7.1	33.3	33.3
Pm+Pl+Q (Total)		84.7	95.1	159.7	106.5	83.8	94.2	160.3	107.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.45	138.00	Passed
Pm+Pl (SUS)	33.28	207.00	Passed
Pm+Pl+Q (TOTAL)	160.31	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: S18"

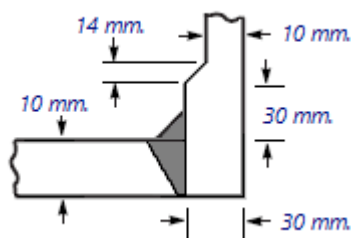
From: 30

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	155	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	155	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	550.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	10.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	150.0000	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	30.0000	mm.
Height of Beveled Transition	L'	14.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: S18"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	8.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $3.5 \cdot 556 \cdot 0.99 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 3.5)$
= 0.6953 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $3.5 \cdot 105 / (138 \cdot 1 - 0.6 \cdot 3.5)$
= 0.2657 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.6833 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 20/0.5773
= 34.6410 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

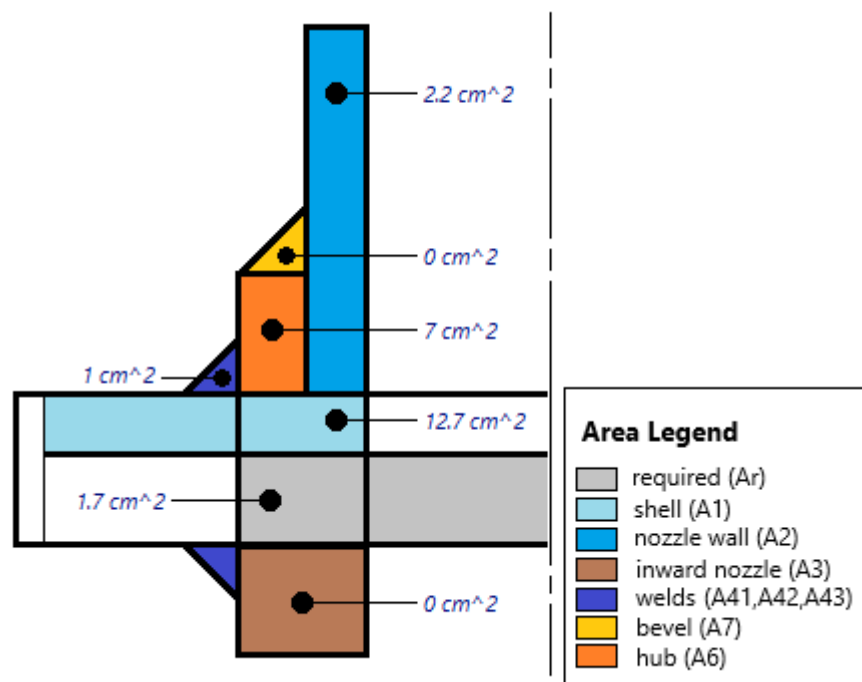
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	458.6887	mm.
Parallel to Vessel Wall, opening length	d	229.3443	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm.

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 138/138)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 138/138)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1, 1)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1.595	1.696	NA
Area in Shell	A1	14.459	12.663	NA
Area in Nozzle Wall	A2	2.365	2.218	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	7.000	7.000	NA
TOTAL AREA AVAILABLE	Atot	24.824	22.881	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 65.81 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F (1 - fr1))$ per UG-37(d)
 $= 0.5 (229 \cdot 1.48 \cdot 1 + 2 \cdot 27 \cdot 1.48 \cdot 1 (1 - 1))$
 $= 1.696 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 229 (1 \cdot 7 - 1 \cdot 1.48) - 2 \cdot 27 (1 \cdot 7 - 1 \cdot 1.48) \cdot (1 - 1)$
 $= 12.663 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$
 $= (2 \cdot 17.5) (7 - 0.68) / \sin(85.4)$
 $= 2.218 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$

= 1.000 cm²

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(17.5, 200, 30)) \cdot (30 - 10) \cdot 1$$

$$= 7.000 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 10, tr = 0.27, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.038, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 10, tr = 0.7, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.099, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 3.50/51.00 = 0.069

Weld Size Calculations, Description: S18"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (1.7 - 12.7 + 2 \cdot 27 \cdot 1 \cdot (1 \cdot 7 - 1.48)) \cdot 138)$$

$$= \max(0, -11237.32) \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - C_{an} / .707)^2 \cdot fr_2) \cdot S_v$$

$$= (9.22 + 1 - 0 \cdot 1) \cdot 138$$

$$= 14378.36 \text{ Kgf}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (9.22 + 0 + 1 + (3.78)) \cdot 138$$

$$= 19697.41 \text{ Kgf}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (9.22 + 0 + 1 + 0 + (3.78)) \cdot 138$$

= 19697.41 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 289 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 31251. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 129) \cdot (30 - 3) \cdot 0.7 \cdot 138$$

$$= 108176. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 289 \cdot (10 - 3) \cdot 0.74 \cdot 138$$

$$= 33037. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (31251 + 108176) = 139427 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (31251 + 0 + 33037 + 0) = 64289 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (31251 + 33037 + 0) = 64289 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 139427 Kgf, must exceed W = 0 Kgf	or W1 = 14378 Kgf
Path 2-2 = 64288 Kgf, must exceed W = 0 Kgf	or W2 = 19697 Kgf
Path 3-3 = 64288 Kgf, must exceed W = 0 Kgf	or W3 = 19697 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 28.2 bars

Nozzle is O.K. for the External Pressure: 1.03 bars

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: S18"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	8.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 3.5 \cdot 556 \cdot 0.99 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 3.5)$$

$$= 0.6953 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 3.5 \cdot 105 / (138 \cdot 1 - 0.6 \cdot 3.5)$$

$$= 0.2657 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.6833 mm.

Intermediate Hub Nozzle Calculations:

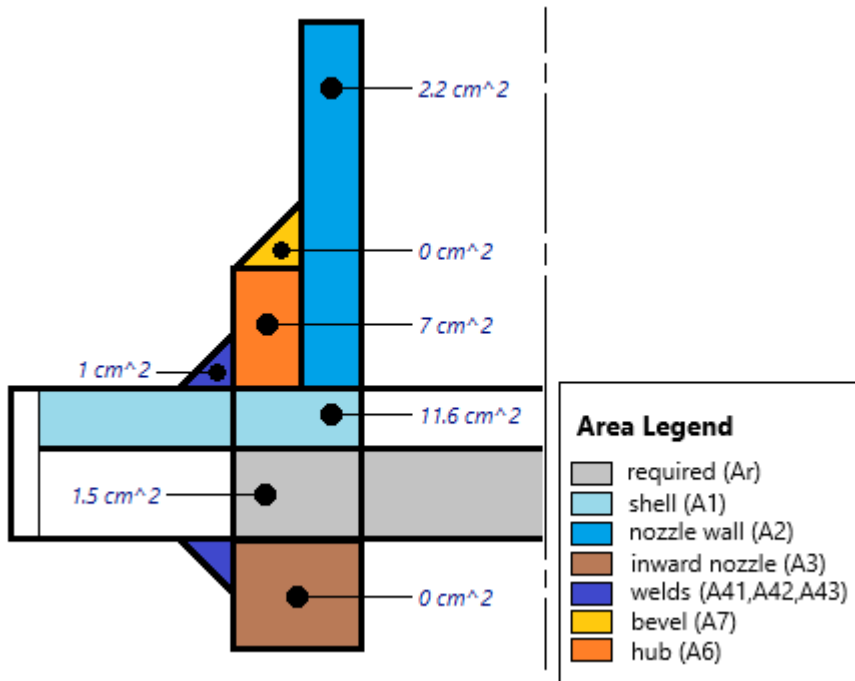
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 20/0.5773$
 $= 34.6410 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 418.4000 mm.
 Parallel to Vessel Wall, opening length d 209.2000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 17.5000 mm.



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1.455	1.547	NA
Area in Shell	A1	13.189	11.551	NA
Area in Nozzle Wall	A2	2.357	2.211	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	7.000	7.000	NA
TOTAL AREA AVAILABLE	Atot	23.546	21.762	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(209 \cdot 1.48 \cdot 1 + 2 \cdot 27 \cdot 1.48 \cdot 1(1-1))$
 $= 1.547 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 209(1 \cdot 7 - 1 \cdot 1.48) - 2 \cdot 27$

$$(1 \cdot 7 - 1 \cdot 1.48) \cdot (1 - 1)$$

$$= 11.551 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2}$$

$$= (2 \cdot 17.5) (7 - 0.68) \cdot 1$$

$$= 2.211 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_{an}/0.707)^2 \cdot f_{r2}$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(17.5, 200, 30)) \cdot (30 - 10) \cdot 1$$

$$= 7.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.6833 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 3.6953 \text{ mm.}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 4.5000 \text{ mm.}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 11.1026 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.1, \max(4.5, 4.5)]$$

$$= 4.5000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.68, 4.5)$$

$$= 4.5000 \text{ mm.}$$

Available Nozzle Neck Thickness = 10.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	33.5,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	311.5 N./mm ²	Passed
Occasional	:	2.5,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	22.9,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: S18"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 9.0000 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) S_v)$$

$$= \max(0, (1.55 - 11.6 + 2 \cdot 27 \cdot 1 \cdot (1 \cdot 7 - 1.48)) 138)$$

$$= \max(0, -9881.79) \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (9.21 + 1 - 0 \cdot 1) \cdot 138$$

$$= 14368.25 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (9.21 + 0 + 1 + (3.78)) \cdot 138$$

$$= 19687.31 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (9.21 + 0 + 1 + 0 + (3.78)) \cdot 138$$

$$= 19687.31 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 263 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 28507. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 118) \cdot (30 - 3) \cdot 0.7 \cdot 138$$

$$= 98674. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 263 \cdot (10 - 3) \cdot 0.74 \cdot 138$$

$$= 30135. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (28507 + 98674) = 127181 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (28507 + 0 + 30135 + 0) = 58642 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (28507 + 30135 + 0) = 58642 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 127180 Kgf, must exceed W = 0 Kgf	or W1 = 14368 Kgf
Path 2-2 = 58642 Kgf, must exceed W = 0 Kgf	or W2 = 19687 Kgf
Path 3-3 = 58642 Kgf, must exceed W = 0 Kgf	or W3 = 19687 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 29.2 bars

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 63.3277 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 273.7945 mm.

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: S18" :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Spherical
Internal Corrosion Allowance	Cas	3.0000 mm.
Vessel Diameter	Dv	990.000 mm.



Vessel Thickness	Tv	10.000	mm.
Design Temperature	T1	155.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm2
Vessel Hot S.I. Allowable	Smh	138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	ID
Corrosion Allowance for Nozzle	Can	3.0000 mm.
Nozzle Diameter	Dn	203.200 mm.
Nozzle Thickness	Tn	30.000 mm.
Nozzle Material		SA-105
Nozzle UNS Number		K03504
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm2
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm2
Design Internal Pressure	Dp	3.500 bars
Include Pressure Thrust		No

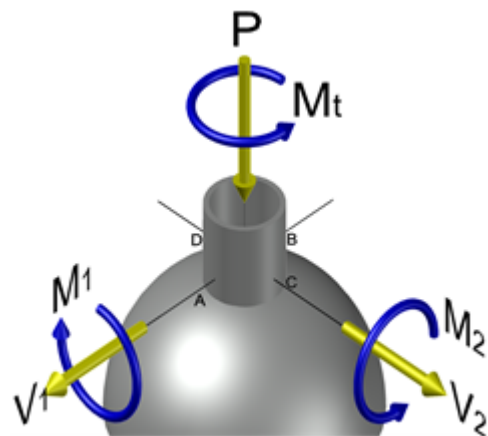
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-1305.3	Kgf
Longitudinal Shear (SUS)	(V1) V1	1153.8	Kgf
Circumferential Shear (SUS)	(Vc) V2	1153.8	Kgf
Circumferential Moment (SUS)	(Mc) M1	605.2	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	605.2	Kg-m.
Torsional Moment (SUS)	Mt	782.9	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 263.2 + 2 \cdot 1.65 \cdot \sqrt{(501.5 (10.0 - 3.0))}$$

= 458.723 mm.

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	-1305.3	Kgf
Circumferential Shear	(VC) V2	1153.8	Kgf
Longitudinal Shear	(VL) V1	1153.8	Kgf
Circumferential Moment	(MC) M1	605.2	Kg-m.
Longitudinal Moment	(ML) M2	605.2	Kg-m.
Torsional Moment	MT	782.9	Kg-m.

Dimensionless Param: U = 2.22 TAU = 5.00 (4.37) RHO = 0.26

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 1	0.01500	(A,B,C,D)
M(x) / P	SP 1	0.01307	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 1	0.01200	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 1	0.01388	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 1	0.01200	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 1	0.01388	(A,B,C,D)
N(y) * T / P	SP 1	0.02325	(A,B,C,D)
M(y) / P	SP 1	0.00507	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 1	0.01475	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 1	0.00414	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 1	0.01475	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 1	0.00414	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	3.9	3.9	3.9	3.9	3.9	3.9	3.9	3.9
Rad. Bend.	P	20.5	-20.5	20.5	-20.5	20.5	-20.5	20.5	-20.5
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-24.5	-24.5	24.5	24.5
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-170.2	170.2	170.2	-170.2
Rad. Memb.	ML	-24.5	-24.5	24.5	24.5	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-170.2	170.2	170.2	-170.2	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-170.3	129.1	219.2	-162.3	-170.3	129.1	219.2	-162.3
Tang. Memb.	P	6.1	6.1	6.1	6.1	6.1	6.1	6.1	6.1
Tang. Bend.	P	7.9	-7.9	7.9	-7.9	7.9	-7.9	7.9	-7.9
Tang. Memb.	MC	0.0	0.0	0.0	0.0	-30.1	-30.1	30.1	30.1
Tang. Bend.	MC	0.0	0.0	0.0	0.0	-50.8	50.8	50.8	-50.8
Tang. Memb.	ML	-30.1	-30.1	30.1	30.1	0.0	0.0	0.0	0.0
Tang. Bend.	ML	-50.8	50.8	50.8	-50.8	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-66.9	18.8	95.0	-22.5	-66.9	18.8	95.0	-22.5
Shear VC		3.9	3.9	-3.9	-3.9	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-3.9	-3.9	3.9	3.9
Shear MT		10.1	10.1	10.1	10.1	10.1	10.1	10.1	10.1
Tot. Shear		14.0	14.0	6.2	6.2	6.2	6.2	14.0	14.0
Str. Int.		172.2	130.9	219.5	162.5	170.7	129.5	220.7	163.7

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm2)



Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5
Rad. Pl (SUS)		-20.6	-20.6	28.4	28.4	-20.6	-20.6	28.4	28.4
Rad. Q (SUS)		-149.7	149.7	190.7	-190.7	-149.7	149.7	190.7	-190.7
Long. Pm (SUS)		12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5
Long. Pl (SUS)		-24.1	-24.1	36.2	36.2	-24.1	-24.1	36.2	36.2
Long. Q (SUS)		-42.9	42.9	58.8	-58.8	-42.9	42.9	58.8	-58.8
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		3.9	3.9	-3.9	-3.9	-3.9	-3.9	3.9	3.9
Shear Q (SUS)		10.1	10.1	10.1	10.1	10.1	10.1	10.1	10.1
Pm (SUS)		12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5
Pm+Pl (SUS)		14.2	14.2	50.3	50.3	14.2	14.2	50.3	50.3
Pm+Pl+Q (Total)		159.8	143.3	231.9	150.1	158.3	141.9	233.2	151.2

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.45	138.00	Passed
Pm+Pl (SUS)	50.30	207.00	Passed
Pm+Pl+Q (TOTAL)	233.16	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

- ** Warning: One or more nozzles do not meet proper Spacing Requirements!**
- ** Warning: A standard flange seems to be governing the Vessel MAWP!**

Please review all reports carefully!

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: